

SECTION C: TECHNICAL REGULATIONS

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[Red Text]:	Information on applicable Governance and relevant Advisory Committee
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ARTICLE C1: GENERAL PRINCIPLES

Advisory Committee: TAC and PUAC

Governance: F1 Commission / PU Manufacturers' Governance Agreement / WMSC

C1.1 Amendments to these Technical Regulations

Any amendments to these Technical Regulations that do not relate specifically to the Power Unit or that relate to matters of form rather than substance (such as re-numbering, reference corrections, etc.) will not be subject to the approval of the PU Manufacturers. Any amendments to these Technical Regulations that relate to substantive matters concerning the Power Unit will be subject to the prior approval of the PU Manufacturers in accordance with the 2026 F1 PU Governance Agreement, as referenced in Article 1.1 of Appendix A7.

C1.2 Dangerous construction

The stewards may prohibit the participation of a vehicle whose construction is deemed to be dangerous. Should the relevant information become apparent during a session, such a decision may apply with immediate effect.

C1.3 Compliance with the regulations

Formula 1 Cars must comply with these regulations in their entirety at all times during a Competition.

Should a Competitor or PU Manufacturer introduce a new design or system or feel that any aspect of these regulations is unclear, clarification may be sought from the FIA Formula One Technical Department. If clarification relates to any new design or system, correspondence must include:

- a. A full description of the design or system.
- b. Drawings or schematics where appropriate.
- c. The Competitor or PU Manufacturer's opinion concerning the immediate implications on other parts of the car of any proposed new design.
- d. The Competitor or PU Manufacturer's opinion concerning any possible long-term consequences or new developments which may come from using any such new designs or systems.
- e. The precise way or ways in which the Competitor or PU Manufacturer feels the new design or system will enhance the performance of the car.

C1.4 New systems or technologies

Any new system, procedure or technology not specifically covered by these Technical Regulations, but which is deemed permissible by the FIA, will only be admitted until the end of the Championship during which it is introduced. Following this the Formula One Commission will be asked to review the technology concerned and, if they consider (in their sole discretion) that such new system, procedure, or technology adds no value to Formula One in general, it may be specifically prohibited by the FIA.

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C1.5 Duty of Competitor and PU Manufacturer

It is the duty of each Competitor to satisfy the FIA and the stewards that its Formula 1 Car complies with these regulations in their entirety at all times during a Competition. With regard to PUs used on a Formula 1 Car, this duty and responsibility also extends to the PU Manufacturer.

The design of the car, its components and systems shall, except for safety features, demonstrate their compliance with these regulations by means of physical inspection of hardware or materials. Unless explicitly requested by an Article, no mechanical design may rely upon software inspection as a means of assessing compliance.

To further demonstrate compliance with Articles C3.2.2 and C3.18.1, Competitors may be required to submit calculations, or to run with analytic devices specified by the FIA.

Due to their nature, the compliance of electronic systems may be assessed by means of inspection of hardware, software, and data.

CAD models may be requested by the FIA to check compliance with the Regulations. Such models should be supplied in a format and by a method specified by the FIA. In such cases, scanning technology will be used by the FIA to check that the physical car is the same as the inspected CAD models.

Each Competitor and each PU Manufacturer must ensure that all relevant personnel (whether employee, consultant, contractor, secondee or any other type of permanent or temporary personnel) associated with its participation in the Championship are appropriately informed with respect to the ways in which their areas of responsibility may impact the compliance of the Competitor and/or PU Manufacturer (as applicable) with the Regulations.

Each Competitor and each PU Manufacturer must ensure that the FIA ethics and compliance hotline with respect to the Regulations is clearly communicated to all relevant personnel.

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ARTICLE C2: CONVENTIONS AND FUNDAMENTAL DIMENSIONS

Advisory Committee: TAC

Governance: F1 Commission / WMSC

C2.1 Coordinate systems and conventions

C2.1.1 Car Coordinate System

The “**Car Coordinate System**” is a right-handed Cartesian [X, Y, Z] coordinate system and is defined in the following way:

- The X axis is the longitudinal direction of the car and increases rearwards.
- The Y axis is the transverse direction of the car and increases to the (driver’s) right-hand side.
- The Z axis is the vertical direction and increases upwards.

C2.1.2 Further conventions

- If no units are specified, it is implicit the unit will be in millimetres
- The terms “inboard” or “outboard”, when used in reference to the Y coordinate, respectively refer to closer to or further away from the plane $Y=0$.
- A suffix may be used for local coordinates in specific rules, (e.g. X_w , Y_w , Z_w), where these local axes are defined within a specific Article for local use.
- Planes will be referred to as the axis to which they are normal to (e.g. X-Plane or $X_A = 300$ plane).
- A wheel is defined to be in the straight-ahead position when its rotational axis lies in an X-Plane.

Unless otherwise specified:

- the positive side of the Y axis is used in the various Articles and it is implicit that a symmetrical rule applies for the other side of the car.
- any measurements and references will be with the wheels in the straight-ahead position.
- when a viewing direction is stated, “front” and “rear” are parallel to the X axis, “side” is parallel to the Y axis (in the direction towards the plane $Y = 0$) and “above”, “below” and “plan” are parallel to the Z axis.
- directions of angles, slopes and incidences are taken in context of the right-handed Cartesian coordinate system defined in C2.1.1. For example, a positive slope within a Y-Plane would be characterised by positively increasing X and Z components.
- references to axis aligned geometry are aligned to the Car Coordinate System defined in C2.1.1.
- a load vector ([x, y, z]kN) will have components parallel to the axes of the Car Coordinate System.

C2.1.3 Wheel Coordinate System

The “**Wheel Coordinate System**” is a local right-handed Cartesian [X_w , Y_w , Z_w] coordinate system and is defined in the following way for each wheel:

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- a. The origin of the Wheel Coordinate System is the intersection of the rotational axis of the wheel and the inboard plane of the Wheel Rim.
- b. The X_W axis lies in the inboard plane of the Wheel Rim and increases in the rearward direction. With the wheel in the straight-ahead position and the car at Legality Setup, the X_W axis is parallel to the car's X-Axis.
- c. The Y_W axis is coincident with the wheel's axis of rotation and increases towards the plane $Y=0$. Referring to this axis, the terms "inboard" or "outboard" respectively refer to closer to or further away from the plane $Y = 0$.
- d. The Z_W axis lies in the inboard plane of the Wheel Rim and increases upwards.
- e. Once the Wheel Coordinate System is defined as above, then it maintains a fixed orientation relative to the suspension upright at all other suspension articulations.

C2.2 Principal Planes

- a. The Plane $Z = 0$ is defined as a horizontal plane sitting at the bottom of the sprung part of the car, except for the Plank Assembly defined in Article C3.6.
- b. The Plane $Y = 0$ is defined as the plane of symmetry of the car.
- c. The Plane $X_A = 0$ is defined as the X-Plane that lies on the forward limit of the Survival Cell.
- d. The Plane $X_C = 0$ is defined as the X-Plane at the rear of the Cockpit.
- e. The planes $X_F = 0$ and $X_R = 0$ are defined as the X-Planes which respectively pass through the origin of the two front or two rear Wheel Coordinate Systems, with the wheels in the straight-ahead position and the car at Legality Setup.
- f. The plane $X_{DIF} = 0$ is defined as the X-Plane containing the axis of rotation of the final drive as defined in Article C9.5.1.
- g. The plane $X_{PU} = 0$ is defined as the X-Plane which passes through the mounting face of the connections between the ICE and the Survival Cell, as defined in Article C5.4.17.
- h. The plane $X_{FIS} = 0$ is defined as the X-plane passing through the forward most point of the Front Impact Structure.

C2.3 Fundamental Dimensions

C2.3.1 Width

Except for the tyres, the Wheel Rims, and parts attached to the Wheel Rims as defined in Article C10.7.6, no part of the car may lie more than 950mm from the plane $Y=0$ at Legality Setup.

C2.3.2 Height

Except for the Plank Assembly, no part of the car may lie directly below the Trim and Combination of the following Reference Volumes:

- a. RV-FLOOR-BODY.
- b. RV-FLOOR-BIB.
- c. RV-FLOOR-FOOT.

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d. RV-FLOOR-SIDEWALL.

e. RV-FLOOR-FENCE.

C2.3.3 Wheelbase

The distance between the planes $X_F = 0$ and $X_R = 0$ must be less than or equal to 3400mm at Legality Setup.

C2.3.4 Front Wheel Position

The plane $X_F = 0$ must lie between $X_A = 0$ and $X_A = 150$ inclusive.

C2.3.5 Cockpit Position

The distance between $X_A = 0$ and $X_C = 0$ must be greater than or equal to 1830mm and less than or equal to 2030mm.

C2.3.6 Rear Bulkhead Position

The distance between $X_C = 0$ and $X_{PU} = 0$ must be greater than or equal to 360mm.

C2.3.7 Exhaust Exclusion Zone

Except for Tailpipe, no part of the car may lie within a right circular cylinder which intersects the planes $X_R = 385$ and $X_{DIF} = 800$, and whose axis is identical to, and diameter 20mm greater than, that of the right circular cylinder defined in C3.9.2(g).

C2.4 Reference Volumes and Surfaces

“Reference Volumes” and “Reference Surfaces” and their position in space are defined in Appendix C2 using the Car Coordinate System and are used throughout the Regulations for geometrical constraints. For convenience, Reference Volumes are preceded by the prefix “RV–” and Reference Surfaces by the prefix “RS–”.

C2.5 Precision of Numerical Values

Any numerical values specified in these Regulations as limits (maxima or minima), will be considered to be the limits regardless of the decimals quoted.

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ARTICLE C3: AERODYNAMIC COMPONENTS

Advisory Committee: TAC

Governance: F1 Commission / WMSC

C3.1 Aerodynamic Components or Bodywork

“Aerodynamic Components” or **“Bodywork”** are parts of the car in contact with the External Air Stream.

- a. The following components are Bodywork:
 - i. all components described in Article C3.
 - ii. inlet or outlet cooling ducts, up to the component they provide cooling for.
 - iii. inlet ducts for the Power Unit (air boxes) up to the air filter.
 - iv. primary heat exchangers, as defined in Article C7.4.1 (b).
- b. The following components are not Bodywork:
 - i. cameras and camera housings, as defined in Article C8.16.
 - ii. rear view mirrors as defined in Article C14.2.
 - iii. the ERS status light.
 - iv. parts associated with the mechanical functioning of the Power Train, transmission of power to the wheels, and the steering system, provided none are designed to achieve an aerodynamic effect.
 - v. Wheel Rims and tyres.
 - vi. brake disc assemblies, calipers, and pads.
 - vii. CCU Antenna

C3.2 General Principles

C3.2.1 Objectives of Article C3

The primary objectives of Article C3 are:

- a. to promote close racing by minimising the aerodynamic performance loss when one car follows another.
- b. to ensure aerodynamic performance remains compatible with the performance of the PU.

To assess these objectives, F1 Teams may be asked to provide relevant information to the FIA upon request.

The Intellectual Property of this information remains with the F1 Team, will be protected, and will not be disclosed to any third parties.

C3.2.2 Aerodynamic Influence

Except for the driver-adjustable bodywork specified in Articles C3.10.10 and C3.11.6 when in a State of Deployment only, minimal parts related to their operation, and Aerodynamic and flexible

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seals allowed by Articles C3.10.11, C3.14, C3.16.3 and C3.17.7, all Aerodynamic Components or Bodywork must:

- a. be rigidly constructed and immobile relative to their defined Frame of Reference defined in Article C3.4.
- b. provide a uniform, solid, hard, continuous, and impervious surface at all times.

Where an Aerodynamic Component complies with a test defined in Article C3.18, it will be considered compliant with (a) above for the deflection tested.

Any device or structure designed to bridge the gap between the sprung part of the car and the ground is strictly prohibited.

Apart from necessary parts for the adjustments described in Articles C3.10.10 and C3.11.6, and incidental movements from the steering system, any car system, device, or procedure that alters the aerodynamic characteristics of the car through driver movement is forbidden.

Furthermore, the aerodynamic effect of any component not classified as Bodywork must be incidental to its primary function. Designs intended to enhance any such aerodynamic effects are prohibited.

C3.2.3 Symmetry

All Bodywork must be nominally symmetrical with respect to $Y = 0$. Therefore, unless otherwise specified, any regulation in Article C3 concerning one side of the car will be assumed to be valid for the other side of the car including references to the maximum number of components allowed per side. Furthermore, any Aerodynamic Surfaces intersecting $Y = 0$ must be Tangent Continuous across this plane.

Minimal exceptions to the requirement of symmetry of this Article will be accepted for the installation of non-symmetrical mechanical components, for asymmetrical cooling requirements or for asymmetrical angle adjustment of the FW Primary and Secondary Flap, provided this adjustment is not controlled by the FIA Standard ECU.

Bodywork on the Unsprung Mass must respect this Article when the suspension position of each wheel is virtually reorientated so that its Wheel Coordinate System axes are parallel to their respective axis of the Car Coordinate system.

C3.2.4 Component Bodywork

Unless otherwise stated, all individual Bodywork Components described in Articles C3.5, C3.7 to C3.11, and Article C3.14, prior to any Trim and Combination operations, and before the addition of Apertures, must:

- a. be single volumes that are simply connected.
- b. in any X, Y and Z plane, or any X_w , Y_w and Z_w plane for Wheel Bodywork, only contain a single section.

C3.2.5 Trim and Combination

A “Trim and Combination” operation can only be performed once the parts to be trimmed are fully defined. The only regions of a volume that can be removed as part of a trimming process are those

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that are internal to the body they are trimmed by. Once parts are trimmed and combined, and any Fillet Radius or Edge Radius permitted by the relevant Article is applied, the resultant volume must maintain both continuity and tangency in any X, Y, and Z plane at the boundaries between the adjacent sections of the parts. Unless otherwise stated:

- a. The elective trimming of volumes beyond or outside of any overlap is not permitted.
- b. Trim and Combination operations must result in a single volume.

C3.2.6 Fillet and Edge Radius

A “Fillet Radius” is formed by rounding an internal corner (included angle less than 180 degrees) with a concave surface by only adding material, whilst an “Edge Radius” is created by smoothing an external corner (included angle greater than 180 degrees) with a convex surface by only removing material.

In both instances, the resulting surface must consist of arcs that adhere to specified radius limits, connect two fully defined surfaces tangentially without inflection, and align perpendicularly to their intersection. Unless otherwise stated, both Fillet and Edge Radii may vary in size along the boundary perimeter, but these changes must be continuous.

If a discontinuity in tangency exists at the trailing edge where a Fillet Radius has joined parts, a Closed Aerodynamic Fairing may be added immediately behind the trailing edge. This fairing must:

- a. maintain both continuity and tangency with the filleted parts.
- b. fit within an Acute Circular Cone whose:
 - i. base radius is 1.2 times the minimum radius necessary to fully enclose the preceding Fillet Radii and any trailing edge immediately adjacent to the Fillet Radii.
 - ii. height is three times the maximum trailing edge Fillet Radius.
- c. comply with Article C3.2.4 (a).

C3.2.7 Pressure Tappings

Pressure measuring apertures are permitted on the car, provided they:

- a. have an internal diameter of no more than 2mm.
- b. are flush with the underlying geometry.
- c. are only connected to pressure sensors or are completely blanked.

C3.2.8 Section and Article Titles

Section and Article Titles within Article C3 have no regulatory value.

C3.2.9 Tangency Continuity

Unless otherwise stated, the Aerodynamic Surfaces of all Bodywork and sub-bodywork components described in C3.5 to C3.11, and C3.14 to C3.15, must maintain Tangency Continuity.

If two adjacent surfaces are not tangent continuous but can become so by applying an Edge Radius or Fillet Radius of up to 1mm along their boundary, they will be considered Tangent

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Continuous at this boundary, provided the addition of these radii do not contravene any relevant Articles.

C3.3 Legality Checking

C3.3.1 Digital legality checking

Assessment of geometric compliance with Article C3 will be carried out digitally using CAD models provided by the teams. In these models:

- a. Components may only be designed to the edge of a Reference Volume, or with a precise geometrical feature, or to the limit of a geometrical criterion (save for the normal round-off discrepancies of the CAD system), when the regulations specifically require an aspect of the bodywork to be designed to this limit, or it can be demonstrated that the design does not rely on lying exactly on this limit to conform to the regulations, such that it is possible for the physical bodywork to comply.
- b. Components that must follow a precise shape, surface or plane, must be designed without any tolerance, save for the normal round-off discrepancies of the CAD system.
- c. Bodywork required to be visible from a prescribed direction may include surfaces parallel to that direction, provided it can be shown that such parallel surfaces could instead be drawn at an infinitesimally small, included angle and still comply with all relevant Articles.

C3.3.2 Physical legality checking

Cars may be measured during a Competition to check their conformance to the CAD models discussed in Article C3.3.1 and/or to ensure they remain inside the Reference Volumes. All measurement will be made relative to the coordinate system determined by the datum points required by C3.3.3.

- a. Unless otherwise stated, a tolerance of $\pm 3\text{mm}$ will be accepted for manufacturing purposes only with respect to the CAD surfaces. Where measured surfaces lie outside of this tolerance but remain within the Reference Volumes, an F1 Team may be required to provide additional information (e.g. revised CAD geometry) to demonstrate compliance with the regulations. Any discrepancies contrived to create a special aerodynamic effect or surface finish are not permitted.
- b. Irrespective of (a), geometrical discrepancies at the limits of the Reference Volumes must be such that the measured component remains inside the Reference Volume.
- c. A positional tolerance of $\pm 2\text{mm}$ will be accepted for the Front Wing Bodywork, Rear Wing Bodywork, Exhaust Tailpipe, Floor Bodywork behind $X_R = -335$, and Tail. This will be assessed by realigning each of the groups of Reference Volumes and Reference Surfaces that define the assemblies, by up to 2mm, from their original position, to best fit the measured geometry.
- d. Irrespective of (b), a tolerance of $Z = \pm 2\text{mm}$ will be accepted for parts of the Floor Bodywork forward of $X_R = -335$ and Plank Assembly, visible from below.
- e. Minimal discrepancies from the CAD surfaces will also be accepted in the following cases:
 - i. Minimal repairs carried out on Aerodynamic Components and approved by the FIA.

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- ii. Tape, provided it does not achieve an aerodynamic effect not otherwise permitted by Article C3.
- iii. Junctions between Bodywork panels.
- iv. Local Bodywork fixing details.

C3.3.3 Datum Points

All cars must be equipped with optical target mountings in the following locations. Unless otherwise stated, all positions have two targets positioned symmetrically about $Y = 0$.

All datum points must be accessible without removing bodywork. It is acceptable to cover the points with tape or with quickly removable covers.

- a. On the top of the Survival Cell, more than 80mm from $Y=0$, and between $X_A = 0$ and $X_A = 300$.
- b. On the top of the Survival Cell, *more than 140mm from $Y = 0$, and* close to $X_C = -875$.
- c. On the side of the Survival Cell with $X_A < 1200$.
- d. On the side of the Survival Cell close to the rear mounts of the Secondary Roll Structure.
- e. Within an axis-aligned cuboid with an interior diagonal defined by points $[X_C = 0, -40, 1120]$ and $[X_C = 150, -175, 840]$.
- f. On the lower surface of the Survival Cell, symmetrically about $Y = 0$, between $X_A = 0$ and $X_A = 500$, and more than 10mm from $Y = 0$.
- g. On the lower surface of the Survival Cell, symmetrically about $Y = 0$, between $X_C = -300$ and $X_C = 100$, and more than 150mm from $Y = 0$.
- h. A single probed point on the RIS or Gearbox Case.

In all cases, a file containing the coordinates of the required datum points must be supplied for each Survival Cell.

Full details of the requirements are given in the document [FIA-F1-DOC-007](#).

C3.3.4 Supports for Scrutineering

All cars must be equipped with 4 pads of diameter 20mm that will be used to support the car during scrutineering. The pads:

- a. must be rigidly mounted to the Survival Cell, ICE or Gearbox Case as appropriate.
- b. must be on a Z-Plane between $Z=2$ and $Z=10$ and will be used to determine the location of the plane $Z = 0$ when inspecting the underside of the car.
- c. must have an M8x1.25 hole in the centre with at least 4 complete threads.
- d. must be in the following locations:
 - i. two positioned symmetrically about $Y = 0$, at the same X-Plane as the centre of the central hole in the plank assembly, defined in C3.6.1 (c) and 150mm from $Y = 0$.
 - ii. two positioned symmetrically about $Y = 0$ at the same X-plane as the centre of the rearmost hole in the plank assembly, defined in C3.6.1 (c) and 105mm from $Y = 0$.

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Details of the mounting requirements are given in the document [FIA-F1-DOC-007](#).

C3.4 Component Definition

The permitted bodywork and relevant Frame of Reference for each group is defined in the following Articles.

C3.4.1 Bodywork part of the sprung mass of the car

The only permissible Sprung Mass Bodywork is defined under Articles C3.5 to C3.13 and Articles C3.1.1 (a) (ii) to (iv). The Frame of Reference for all Sprung Mass Bodywork is the Car Coordinate System.

Any Bodywork subject to Final Assembly as per Article C3.12, must first be classified under one of the groups specified in Articles C3.5 to C3.11.

Compliance of each Bodywork *component* with Article C3 will be assessed independently and before any Trim and Combination operations and Fillet application described in Article C3.12, and any Aperture application described in Article C3.16.

Compliance of the Aerodynamic Surfaces of each Bodywork component with Article C3 will be assessed independently and after their boundaries have been defined by Trim and Combination operations and Fillet application described in Article C3.12, and before any Aperture application described in Article C3.16.

Furthermore, the use of components not defined as Bodywork to achieve compliance with the Bodywork regulations is not permitted. The FIA may request to inspect any removed geometry post-assembly.

After final assembly, provided any modification does not result in the surfaces becoming Aerodynamic Surfaces, modifications are allowed to the following:

- a. Ducts (as specified in C3.1 (a) (ii) and C3.1 (a) (iii).
- b. Non-Aerodynamic Surfaces.

C3.4.2 Wheel Bodywork

The only permissible Wheel Bodywork is defined in Articles C3.14 and C3.15. The Frame of Reference for all parts classified as Wheel Bodywork is the Wheel Coordinate System.

C3.4.3 Suspension Fairings

The only permissible Suspension Fairings are defined in Article C3.17. To assess compliance with Article C3.2.2, the Frame of Reference of any Suspension Fairing is the structural suspension member that it is attached to.

C3.4.4 Hanger

“Hanger” Bodywork, where permitted under the relevant Articles, is exempt from Article C3.2.4 (b), but must:

- a. lie in its entirety within an individual, freely positioned instance of RV-HANGER.
- b. be rigidly fixed to any bodywork it intersects.

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- c. within **a the plane of symmetry of RV-HANGER situated between the two largest faces of RV-HANGER**, connect two non-intersecting bodywork sections.

C3.5 Floor Bodywork

C3.5.1 Floor Body

“Floor Body” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BODY.
- b. when viewed from below, fully obscure:
 - i. RV-PU-ICE
 - ii. RV-DIFF.
- c. have up to two sections in any Z-Plane.

C3.5.2 Floor Foot

“Floor Foot” Bodywork must:

- a. lie in its entirety within RV-FLOOR-FOOT.
- b. when viewed from above, fully obscure RS-FLOOR-FOOT.
- c. have up to two sections in any Z-Plane.

C3.5.3 Floor Sidewall

“Floor Sidewall” Bodywork must:

- a. lie in its entirety within RV-FLOOR-SIDEWALL.
- b. have up to two sections in any Y-Plane.
- c. **forward of $X_R = 250$ and** above $Z = 100$, have up to two sections in any Z-Plane.

C3.5.4 Main Floor

“Main Floor” results from the Trim and Combination of the following components:

- a. Floor Body.
- b. Floor Foot.
- c. Floor Sidewall.

A Fillet Radius, no greater than 30 mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Main Floor must:

- d. be a single, simply connected volume, with no overlapping regions.
- e. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.
- f. when viewed from the side, fully obscure RS-FLOOR-SIDEWALL.

Main Floor Aerodynamic Surfaces must:

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- g. contain no radius of curvature less than 25mm, except for regions:
 - i. within 5mm of the plan-view-boundary of Main Floor, when viewed from above or below.
 - ii. within 45mm of a single point that lies in RV-FLOOR-SPHERE.
 - iii. of convex curvature, visible from above, and within 10mm of the load application points defined in Article C3.18.6.
 - iv. within 5mm of the Fillet Radii permitted as part of the Main Floor Trim and Combination, where the radius of curvature must be no less than 5mm.

C3.5.5 Floor Board

“Floor Board” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BOARD.
- b. have up to three sections in any X, Y and Z-Plane.

Floor Board Aerodynamic Surfaces visible from the side must:

- c. rearward of $X_F = 825$, at any point, in any Z-Section have a:
 - i. normal with a positive X component.
 - ii. tangent that forms an angle of at least 15 degrees to the X-Axis.
- d. contain no concave radius of curvature less than 100mm.

Furthermore:

- e. compliance with (c) and (d) is not required within 35mm of up to three points, which must sit forward of $X_F = 1100$ and be no closer than 50mm to each other.

C3.5.6 Floor Bib

“Floor Bib” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BIB.
- b. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.

Floor Bib Aerodynamic Surfaces must:

- c. contain no concave radius of curvature:
 - i. visible from below.
 - ii. less than 50mm, if visible from above.

C3.5.7 Floor Leading Edge Device

“Floor Leading Edge Device” Bodywork must:

- a. lie in its entirety within RV-FLOOR-LED.
- b. have up to ~~five~~ **three** sections in any X and Z-Plane.

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- c. above $Z = 60$, be no less than 4mm thick when measured normal to Floor Leading Edge Device Aerodynamic Surfaces only.

Floor Leading Edge Device Aerodynamic Surfaces above $Z = 60$ must:

- d. contain no convex radius of curvature less than 2mm

C3.5.8 Floor Winglet

“Floor Winglet” Bodywork, which is exempt from Article C3.2.4, must be fitted and results from the Trim and Combination of the following components:

- a. RS-FLOOR-WINGLET.
- b. **“Winglet Profile”**, which must:
 - i. lie in its entirety within RV-FLOOR-WINGLET.
 - ii. contain no more than a single section in any X, Y and Z-Plane.
- c. A single Hanger, which must:
 - i. when viewed from below, be fully obscured by RS-FLOOR-WINGLET.

Furthermore, Winglet Profile Aerodynamic Surfaces must:

- d. contain no concave radius of curvature less than 20mm.

Once Trimmed and Combined, Floor Winglet and the annotated surface ‘*TRIM-SURFACE*’ may only be modified together, and in the following order:

- e. Rotated by ± 5 Degrees about a Y-aligned axis that passes through the point $[X_R, Z] = [295, 200]$
- f. Translated by ± 10 mm in Z.

Furthermore, Floor Winglet may be:

- g. trimmed below the annotated surface ‘*TRIM-SURFACE*’ in RS-FLOOR-WINGLET.

Once fully defined, Floor Winglet must:

- h. have up to two sections in any Z-Plane below the annotated surface ‘*TRIM-SURFACE*’ in RS-FLOOR-WINGLET.
- i. ~~except for C3.5.8(c),~~ lie in its entirety within RV-FLOOR-WINGLET-ENVELOPE.

C3.5.9 Floor Fence

“Floor Fence” Bodywork must:

- a. lie in its entirety within:
 - i. an axis aligned cuboid of the following dimensions $[X, Y, Z] = [450, 40, 200]$
 - ii. RV-FLOOR-FENCE.
- b. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.
- c. have up to two sections in any Y-Plane.

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Furthermore,

- d. when considering a section formed by the intersection with any Z-Plane, within 100mm of the rearmost point of the section, a line tangent to any part of the section visible from the side must not form an angle greater than 5 degrees, when measured against the X-axis.

C3.5.10 Floor Corner

“Floor Corner” Bodywork is exempt from Article C3.2.4, but must:

- a. lie in its entirety within RV-FLOOR-CORNER.
- b. have up to 3 sections in any X-Plane.
- c. have up to 4 sections in any Y-Plane.
- d. have up to 2 sections in any Z-Plane.

C3.5.11 Floor Tyre IR Sensor

“Floor Tyre IR Sensor” Bodywork must:

- a. lie in its entirety within a freely positioned instance of RV-FLOOR-TYRE-IR.
- b. not lie within RV-FLOOR-CORNER.
- c. be fully obscured by RS-FLOOR-BODY when viewed from above.
- d. fully enclose an infra-red sensor for measuring tyre temperature.

Floor Tyre IR Sensor Aerodynamic Surfaces must:

- e. contain no radius of curvature less than 20mm.

C3.5.12 Floor Bodywork Assembly

“Floor Bodywork Assembly” results from the Trim and Combination of the following components:

- a. Main Floor.
- b. Floor Bib.
- c. Floor Leading Edge Device.
- d. Floor Fence.
- e. Floor Board.
- f. Floor Winglet.
- g. Floor Corner.
- h. Floor Tyre IR Sensor, if fitted.

Before trimming, the following must be discarded:

- i. any Floor Fence remaining above Main Floor.
- j. any Floor Board remaining below Main Floor.
- k. any Floor Winglet remaining inboard of Floor Sidewall.
- l. any Main Floor remaining within RV-FLOOR-CORNER.

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- m. any Floor Tyre IR Sensor remaining below Main Floor.

A Fillet Radius, no greater than 25mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Floor Bodywork Assembly must:

- n. be a single volume, with no overlapping regions.
- o. when viewed from below, fully obscure RS-FLOOR-BODY.
- p. when viewed from below, be coincident with the surfaces of RV-FLOOR-BODY that are visible from below and directly beneath:
 - i. RS-FLOOR-REF.
 - ii. RS-FLOOR-STEP.
- q. forward of $X_F = 1500$, when viewed from the rear, fully obscure Floor Leading Edge Device.
- r. when viewed from the side, fully obscure RS-FLOOR-BOARD.

C3.5.13 Floor Bodywork Group

Once the components defined in Articles C3.5.1 to C3.5.12 have been constructed in accordance with these provisions, including any sub-assembly operations, the resulting union is defined as “**Floor Bodywork**”.

C3.5.14 Floor Auxiliary Components

Once Floor Bodywork has been fully defined and Final Assembly in Article C3.12 completed, the following “**Floor Auxiliary Components**”, which are exempt from Article C3.2.4 (b), may be fitted:

- a. Up to three “**Floor Body Stays**”, which may be designed to only take load in tension, and each of which must:
 - i. have a single inboard attachment ~~location point anywhere on the Sprung Mass except Floor Bodywork~~ that lies between $X_F = 1775$ 2500 and $X_R = 275$ 250, inboard of $Y = 400$ and above $Z = 175$.
 - ~~ii. be fixed on its inboard end to the entirely sprung part of the car.~~
 - ii. ~~be fixed on its outboard end to~~ have a single outboard attachment point that lies on Floor Bodywork between $X_F = 2500$ and $X_R = 250$.
 - iii. have a circular cross section and lie in its entirety within a 5mm diameter right circular cylinder, except within 25mm of the inboard and outboard attachment points, or within 10mm of any adjustment mechanism.
 - iv. when viewed from below, be fully obscured with Floor Bodywork in place.
 - v. not intersect any surfaces of Bodywork, except within 25mm of the inboard and outboard attachment points, and within 7mm of up to two points which lie on the Engine Cover.
- b. Up to eight Hangers, each of which must:
 - i. intersect Floor Board.

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- ii. lie no closer than 50mm to any other Hanger.
 - iii. when viewed from the side and below, be fully obscured with Floor Bodywork in place.
 - iv. lie in its entirety within the union of RV-FLOOR-BOARD and RV-FLOOR-FOOT.
- c. A single “**Floor Board Brace**”, which must:
 - i. lie in its entirety within a 40mm diameter right circular cylinder and RV-FLOOR-BRACE, except within 2mm of Forward Chassis and 35mm of the intersection point of the right circular cylinder’s axis and Forward Chassis.
 - ii. be rigidly fixed at its inboard end to Forward Chassis.
 - iii. be rigidly fixed at its outboard end to Floor Board.
 - iv. when viewed from the side, be fully obscured with Floor Bodywork in place, except within 2mm of Forward Chassis and 35mm of the intersection point of the right circular cylinder’s axis and Forward Chassis.
- d. A single fairing, if necessary, to shroud the single device permitted in Article C12.7, which must:
 - i. be fully within RV-BIB-STAY and be symmetrically arranged about $Y = 0$.
 - ii. in any Z-plane contain only one section.
- e. A single “**Floor Foot Stay**”, which may be designed to only take load in tension, must:
 - i. have a single inboard attachment point that lies no more than 25mm distant from the intersection of Floor Board Brace and Forward Chassis, or in the absence of a Floor Board Brace, within RV-FLOOR-BRACE.
 - ii. be fixed at its inboard end to Forward Chassis.
 - iii. be fixed at its outboard end to Floor Foot.
 - iv. when viewed from the side, be fully obscured with Floor Bodywork in place.
 - v. lie in its entirety within a 10mm diameter right circular cylinder, except within 25mm of the inboard and outboard attachment points, or within 10mm of any adjustment mechanism.

Furthermore, once all Floor Auxiliary Components have been defined:

- f. they must be Trimmed and Combined with any Bodywork they intersect.
- g. a Fillet Radius no greater than 20mm, may be applied along the intersections between Floor Auxiliary Components and any other Bodywork 5mm may be applied along the intersections between Floor Body Stays and any other Bodywork. Fillet radii, not greater than 20mm, may be applied along the intersections between any other Floor Auxiliary Component and any other Bodywork.

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C3.6 Plank and Skids

C3.6.1 Plank Assembly

A “**Plank Assembly**” must be fitted, consisting of the Plank, Skids and mountings.

The following provisions apply to the Plank Assembly:

- a. The upper surface of the Plank Assembly must lie at $Z = 0$, and must not be in contact with the External Air Stream.
- b. The geometry of the Plank Assembly must conform to RV-PLANK with a general manufacturing tolerance of $\pm 0.5\text{mm}$ when new, unless a specific dimensional tolerance is defined.
- c. The Plank Assembly must have three 34mm diameter holes whose axes are parallel to the Z-axis. The centres must lie on $Y = 0$ and have X-locations:
 - i. $X_F = 500$ 800,
 - ii. $-600 \geq X_C \geq -800$, and on or behind $X_F = 1000$.
 - iii. $470 \leq X_{PU} \leq 630$. Furthermore, the centre of the rearmost hole must lie and on or ahead of $X_R = -500$.
- d. The thickness of the Plank Assembly measured normal to the lower surface must be $10\text{mm} \pm 0.2\text{mm}$ and must be uniform when new. A minimum thickness of 8mm will be accepted due to wear, and conformity to this provision will be checked at the peripheries of the holes required by (c) above.

Evaluation of Skid wear will be made by physical measurement of the Skid with it removed from the car. The Measurement will be made with a micrometer with a 6.35mm diameter anvil. When measured in this way, the entire periphery of each hole must be within the specified tolerance.

Four additional 10mm diameter holes are permitted provided their sole purpose is to allow access to the bolts which secure the FIA ADR to the Survival Cell.

C3.6.2 Plank

The following provisions apply to the “**Plank**”:

- a. The material of the Plank is defined in C15.3.5. ~~free, but it must be homogeneous with a specific gravity between 1.3 and 1.45, or if pocketed be made from a bonded assembly the upper 0.5mm of which must have a specific gravity of between 1.3 and 1.65 and the remainder, excluding pockets, be made from a homogeneous material with a specific gravity of between 1.3 and 1.45.~~
- b. The Plank may comprise no more than three pieces, the forward one of which may not be any less than ~~900mm~~ 600mm in length.
- c. Between $X_F = 630$ 930 and $X_C = -800$ and behind $X_C = -400$, pocketing of the Plank is permitted between $Z = -7$ and $Z = -0.5$. The periphery of any pocket, in any Z-Plane, must be not less than 10mm from either the edges of the Plank or any holes or recesses in the Plank. In any vertical plane, the internal pocket Fillet Radii must be at least 3mm and in any Z-Plane, must be at least 10mm. Pockets may only be filled with a material having a specific gravity of less than 0.25.

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C3.6.3 Skids

The lower surface of the Plank may be fitted with flush mounted Skid material which can only be fitted in place of plank material. **“Skids”** may only be fitted around each of the 3 holes defined in C3.6.2.c and must conform to the geometries specified below.

A metallic Skid is mandatory around the forward-most Plank hole. A metallic Skid is optional around the middle and rearward-most Plank holes. If a metallic Skid is not present, it may be either; replaced with a separate piece of Plank material, or integrated into the Plank. The resulting geometry must contain the hole and counterbore detail present in RV-SKID-MID and RV-SKID-REAR.

- a. The Skid material around the forward-most hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-FWD and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
 - iii. fastened directly to the Front Floor Structure with zero degrees of freedom
- b. If metallic, the Skid material around the central hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-MID and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
- c. If metallic, the Skid material around the rear-most hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-REAR and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
- d. Metallic Skids must be made from materials as defined in C15.3.6 either:
 - iii. ~~Titanium alloy Ti6Al4V (according to AMS4928 or AMS4911) in annealed condition. No additional heat treatment is permitted; or~~
 - iv. ~~17-4PH stainless steel (to AMS 5604 / ASTM A 693 or AMS 5643/ ASTM A 564). When the skid is new, its hardness must be no more than 38 HRC.~~

~~Furthermore, they may only be machined from solid and no processes (such as forging, rolling, welding, or coating) may be carried out either before or after machining.~~

The material to be used in each Competition is defined in FIA-F1-DOC-041.

The only permitted deviation from the specified geometry is the removal of minimum material for the fasteners.

C3.6.4 Plank and Skid Mountings

The Plank, and Skids must be fixed to the car using fasteners which are no smaller than M6 and are made from grade 12.9 or 10.9 steel.

If used to attach a Skid to the car, the number of fixings for each type of Skid is defined as follows:

- a. RV-SKID-FWD: three fixings in each of the two forwardmost resulting Skids and two fixings for the rearward most resulting Skid.

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- b. RV-SKID: two fixings in each of the resulting Skids.
- c. when viewed from below, no part of the Skid can be more than 50mm from the centre line of a fastener which passes through that Skid.
- d. No fastener machining, including counterbores or countersinks, may lie in the areas indicated in drawings FIA-LEG-0235 and FIA-LEG-0236.

If used to attach the Plank to the car:

- e. May use a load spreading washer if required.
- f. The total area of the fasteners and any load spreading washers employed with them when viewed from below must be less than 10 000 mm², the area of any single fastener plus its load-spreading washer may not exceed 500mm².
- g. no part of any load-spreading washer may be below Z = -7. The Skids referred to in C3.6.3 are not considered as load spreading washers.

The following provisions apply to each single fastener

- h. the shanks of the fasteners (which may be no less than 6mm diameter) must be the weakest point in the attachment of metallic Skids to the car.
- i. no part of any fastener may be below Z = -7.5.

C3.7 Front Bodywork

C3.7.1 Nose

“Nose” Bodywork must:

- a. lie in its entirety within RV-NOSE.
- b. when viewed from above, fully obscure RS-NOSE ahead of $X_A = 0$.

Nose Aerodynamic Surfaces, must be Tangent Continuous, and in any X-Plane must:

- c. contain no concave radius of curvature.
- d. between $X_F = -950$ and $X_A = 0$, have a section that when viewed from above:
 - i. is tangent to the Z-Axis at its outermost extremity.
 - ii. contains no radius of curvature less than 45mm at $X_A = 0$.
 - iii. contains no radius of curvature less than 20mm forward of $X_A = 0$.

Furthermore, the following will be exempt from the above:

- e. Cameras in Position 2.
- f. Mounting brackets defined in Article C8.16.7.

C3.7.2 Forward Chassis

“Forward Chassis” Bodywork must:

- a. lie in its entirety within RV-CH-FRONT.
- b. fully enclose RV-CH-FRONT-MIN.

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- c. have up to two sections in any Z-Plane.

Forward Chassis Aerodynamic Surfaces, in any X-Plane, must contain:

- d. no convex radius less than 45mm.
- e. no concave radius less than 500mm.

C3.7.3 Mid Chassis

“Mid Chassis” Bodywork must:

- a. lie in its entirety within RV-CH-MID.
- b. have up to two sections in any Z-Plane.

Mid Chassis Aerodynamic Surfaces must contain:

- c. no concave radius of curvature less than 50mm, except for regions within:
 - i. 50mm of the Secondary Roll Structure.
 - ii. 25mm of the Aperture defined in Article C3.16.10.
- d. no convex radius of curvature less than 25mm below Z = 200.

C3.7.4 Roll Hoop

“Roll Hoop” Bodywork is exempt from Article C3.2.4, but must:

- a. lie in its entirety within RV-ROLL-HOOP.

C3.7.5 Mirror

“Mirror Body” Bodywork must:

- a. lie in its entirety within RV-MIRROR-BODY.
- b. fully enclose an individual instance of RV-LATERAL-SAFETY-LIGHT which may be freely translated along any axis, and rotated up to +/-5deg about any Z-axis only.
- c. have an external surface coincident with the highlighted surface of RV-LATERAL-SAFETY-LIGHT.

“Mirror Inner Stay” Bodywork must:

- d. lie in its entirety within RV-MIRROR-ISTAY.
- e. intersect Mirror Body and Mid Chassis.

“Mirror Rear Stay” bodywork must:

- f. lie in its entirety within RV-MIRROR-RSTAY.
- g. intersect Mirror Body and Sidepod.
- h. in any X-Plane, measure less than:
 - i. 50mm in Z.
 - ii. 10mm in Y.

“Mirror” results from the Trim and Combination of the following components:

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- i. Mirror Body.
- j. Mirror Inner Stay.
- k. Mirror Rear Stay.

A Fillet Radius, no greater than 10mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Mirror must:

- l. be a single volume with no overlapping regions.

C3.7.6 Driver Cooling

“Driver Cooling” Bodywork must:

- a. lie in its entirety within RV-DRI-COOL.
- b. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.

Driver Cooling Aerodynamic Surfaces must:

- c. contain no radius of curvature less than 10mm.

C3.7.7 Front Bodywork Assembly

“Front Bodywork Assembly” results from the Trim and Combination of the following components:

- a. Nose.
- b. Forward Chassis.
- c. Mid Chassis.
- d. Roll Hoop.
- e. Mirror.
- f. Driver Cooling, if present.

A Fillet Radius, no greater than 15mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Front Bodywork Assembly must:

- g. be a single volume, with no overlapping regions.

C3.7.8 Front Bodywork Group

Once the components defined in Articles C3.7.1 to C3.7.7 have been constructed in accordance with these provisions, including any sub-assembly operations, the resulting union is defined as **“Front Bodywork”**.

C3.8 Rear Bodywork

C3.8.1 Sidepod

“Sidepod” Bodywork must:

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- a. lie in its entirety:
 - i. within RV-SIDEPOD.
 - ii. more than 50mm from Floor Board.

Sidepod Aerodynamic Surfaces must contain:

- b. no convex radius less than 50mm.
- c. no concave radius less than 100mm.

Furthermore, compliance with:

- d. (b) and (c) is not required within 20mm of the upper Side Impact Structure, where the radius of curvature must not be less than 10mm.
- e. (b) is not required within 25mm of the Aperture C3.16.9, where the convex radius must not be less than 5mm.

C3.8.2 Engine Cover

“Engine Cover” Bodywork must:

- a. lie in its entirety within RV-EC.
- b. have up to two sections in any Z-Plane.
- c. when viewed from the side, fully obscure RS-EC.

Engine Cover Aerodynamic Surfaces, in any X-Plane outboard of $Y = 5$, must contain:

- d. no convex radius of curvature less than 75mm.
- e. no concave radius of curvature less than 50mm.

Furthermore, when considering Aerodynamic Surfaces only:

- f. compliance with (d) and (e) is not required:
 - i. within 20mm of the upper and lower Side Impact Structures, where the radius of curvature, measured independent of direction, must not be less than 10mm.
 - ii. within 20mm of the CCU Antenna, defined in Article C8.5.4, where the radius of curvature, measured independent of direction, must not be less than 5mm.
 - iii. inboard of $Y = 25$, where the radius of curvature, in any X-Plane, must not be less than 25mm.
- g. rearward of $X_R = -300$ and below $Z = 350$, the X component of any normal to the surface visible from the side must not be negative.
- h. there must be no surface parallel to an X-Plane ahead of $X_R = -55$.

C3.8.3 Rear Bodywork Assembly

“Rear Bodywork Assembly” results from the Trim and Combination of the following components:

- a. Sidepod.
- b. Engine Cover.

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Once fully defined, Rear Bodywork Assembly must:

- c. be a single, simply connected volume, with no overlapping regions.

C3.8.4 Rear Bodywork Group

Once the components defined in Articles C3.8.1 to C3.8.3 have been constructed in accordance with these provisions, including any sub-assembly operations, the resulting union is defined as **“Rear Bodywork”**.

C3.9 Tail and Exhaust Tailpipe

C3.9.1 Tail

“Tail” Bodywork must:

- a. lie in its entirety within RV-TAIL.
- b. when viewed from below, be fully obscured by Floor Body forward of $X_R = 295$.
- c. below $Z = 450$, have up to three sections in any Z-Plane.

C3.9.2 Exhaust Tailpipe

“Exhaust Tailpipe” Bodywork ~~is exempt from Article C3.2.4 (a), but~~ must:

- a. lie in its entirety within RV-TAILPIPE.

Tailpipe Bodywork is exempt from C3.2.4 (a) and, assessing (b) to (g) across both sides of the car, must:

- b. have up to two sections in any Y and Z-Plane.
- ~~c. have a wall thickness of between 0.5mm and 3mm.~~
- c. only have an exit whose entire circumference lies:
 - i. between $X_R = 390$ and $X_R = 400$.
 - ii. above $Z = 350$.
- d. over its last 370mm, have an internal surface with a circular internal cross-section of a constant diameter between 90mm and 130mm.
- e. lie entirely within 3mm from the internal surface when measured normal to the internal surface and in a direction that does not pass through $Y=0$.
- f. be no less than 0.5mm thick (being the minimum distance when measured normal to the surface).
- ~~g. remain unobstructed internally and in full compliance with the provisions of this Article after Final Assembly of all Bodywork groups and application of Apertures.~~
- ~~When considering both sides of the car, over its last 150mm it must:~~
- ~~h. comprise of a single tailpipe and a support:~~
 - ~~i. The support is exempt from Article C3.2.4 and must lie in its entirety within an individual instance of RV-TAILPIPE-BRACKET.~~

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~~ii. RV-TAILPIPE-BRACKET may have a free orientation in space but must intersect both Exhaust Tailpipe and Tail.~~

- g. over its last 150mm have an internal surface which is a right circular cylinder and has a single axis that:
- lies on $Y = 0$.
 - forms an angle between 0.0 and 2.5° to the X-Axis (tail up).

C3.9.3 Tailpipe Bracket

“Tailpipe Bracket” Bodywork is exempt from C3.2.4 and, when considering both sides of the car, must:

- lie in its entirety within a freely positioned, individual instance of RV-TAILPIPE-BRACKET.
- lie in its entirety between $X_R = 185$ and $X_R = 385$.
- intersect both Tailpipe and Tail.

C3.9.4 Exhaust Tailpipe

“Exhaust Tailpipe” results from the Trim and Combination of the following components:

- Tailpipe.
- Tailpipe Bracket.

Once fully defined, Exhaust Tailpipe must:

- be a single volume, with no overlapping regions.
- remain unobstructed internally and in compliance with the provisions of this Article after Final Assembly of all Bodywork groups and application of Apertures.

C3.10 Front Wing (FW)

C3.10.1 Front Wing Profiles

“Front Wing Profiles” Bodywork is exempt from C3.2.4 (b) but must:

- lie in its entirety within RV-FW-PROFILES.
- comprise of up to three, non-intersecting, simply connected volumes.
- have up to three sections in any Y-Plane.
- when viewed from above, fully obscure RS-FW-PROFILES.

In any Y-Plane:

- the rearmost point of every section must be visible when viewed from below.
- except for the rearmost section, the rearmost point of every section must not be visible when viewed from above.
- assessing each section independently, within 40mm of the rearmost point of each section:
 - a line tangent to any part of the section visible from below must have a positive slope. The slope of this line will be considered in the Y-Plane.

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- ii. no part of the section visible from above may be more than 10mm distant from the section visible from below, if outboard of $Y = 400$, or 15mm if inboard of $Y = 400$.
- h. contain:
 - i. no concave radius of curvature visible from below.
 - ii. no concave radius of curvature less than 50mm visible from above.

Furthermore:

- i. when measured against a vertical plane normal to RS-FW-SECTION, contain no normal to any point on the surface that subtends an angle greater than 30° . For volumes thicker than 0.5mm in their entirety (measured normal to the surface on any Y-Plane), compliance with this Article may be determined prior to application of any Edge Radii necessary to demonstrate compliance with C3.3.1 (a).
- j. compliance with (i) is not required for surfaces that lie exactly on $Y = 0$ or $Y = 675$.
- k. At all points along the span, the minimum distance between adjacent Front Wing Profiles volumes must be between 5mm and 15mm when measured with a spherical gauge.
- l. The rearmost point of every individual Y-section, when projected in Z onto a plane through $Z = 0$, must produce a single tangent continuous curve with no radius of curvature less than 200mm.

Once the Front Wing Profiles bodywork is fully defined, Gurneys up to 10mm in height may be fitted to the trailing edge of the upper surface of the rearmost section. These Gurneys are considered to be part of the Front Wing Profiles and must satisfy the provisions of this Article except for (g) and (h) and, for the inner extremity of the innermost Gurney and outer extremity of the outermost Gurney, (i).

C3.10.2 Front Wing Endplate Body

“Front Wing Endplate Body” Bodywork must:

- a. lie in its entirety within RV-FWEP-BODY.
- b. have up to two sections in any Y-Plane.

Front Wing Endplate Body Aerodynamic Surfaces must contain:

- c. no convex radius of curvature less than 5mm.
- d. no concave radius of curvature less than 100mm.

C3.10.3 Front Wing Outboard Footplate

“Front Wing Outboard Footplate” Bodywork must:

- a. lie in its entirety within RV-FWEP-OFP.
- b. below $Z = 160$, have up to two sections in any Z-Plane.
- c. rearwards of $X_F = -425$, have up to two sections in any X-Plane.

Front Wing Outboard Footplate Aerodynamic Surfaces must contain:

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- d. no radius of curvature less than 5mm.
- e. no concave radius of curvature less than 50mm visible from below.
- f. no concave radius of curvature less than 50mm outboard of Y = 825mm.

C3.10.4 Front Wing Inboard Footplate

“Front Wing Inboard Footplate” Bodywork must:

- a. lie in its entirety within RV-FWEP-IFP.

C3.10.5 Front Wing Endplate Diveplane

“Front Wing Endplate Diveplane” Bodywork must:

- a. lie in its entirety within:
 - i. an axis aligned cuboid of the following dimensions [X, Y, Z] = [300, 325, 50].
 - ii. RV-FWEP-DIVEPLANE.

Front Wing Endplate Diveplane Aerodynamic Surfaces must contain:

- b. no convex radius of curvature less than 5mm.
- c. no concave radius of curvature less than 50mm.

C3.10.6 Front Wing Endplate

“Front Wing Endplate” results from the Trim and Combination of the following components:

- a. Front Wing Endplate Body.
- b. Front Wing Outboard Footplate.
- c. Front Wing Inboard Footplate.
- d. Front Wing Endplate Diveplane, if fitted.

Before trimming, any Front Wing Endplate Body remaining below Front Wing Inboard Footplate must be discarded.

A Fillet Radius, no greater than 40mm, may be applied along the intersections between Front Wing Endplate Body and Front Wing Diveplane. A Fillet Radius, no greater than 10mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Front Wing Endplate must:

- e. be a single, simply connected volume, with no overlapping regions.
- f. be no less than 10mm thick (being the minimum distance when measured normal to the surface), if visible from the side.
- g. when viewed from above, fully obscure RS-FWEP-TOP:
- h. when viewed parallel to the Y-Axis from inboard, fully obscure RS-FWEP-SIDE and Front Wing Endplate Diveplane.

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C3.10.7 Front Wing Pylon

“Front Wing Pylon” Bodywork must:

- a. lie in its entirety within RV-FW-PYLON.
- b. in any Z-Plane:
 - i. have a total area no greater than 6000 mm².
 - ii. have a section thickness no more than 35mm when measured in the Y direction.

C3.10.8 Front Wing Strake

“Front Wing Strake” Bodywork must:

- a. lie in its entirety within:
 - i. an axis aligned cuboid of the following dimensions [X, Y, Z] = [775, 20, 100].
 - ii. RV-FW-STRAKE.
- b. have up to two sections in any Y or Z-Plane.

C3.10.9 Front Wing Assembly

“Front Wing Assembly” results from the Trim and Combination of the following components:

- a. Front Wing Profiles.
- b. Front Wing Endplate.
- c. Front Wing Pylon.
- d. Front Wing Strake, if fitted.

Before trimming is applied, any Front Wing Profiles remaining outboard of Front Wing Endplate and any Front Wing Pylon remaining below Front Wing Profiles must be discarded.

A Fillet Radius, no greater than 10mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Front Wing Assembly must:

- e. be a single volume, with no overlapping regions.
- f. when viewed from below, fully obscure Front Wing Pylon.
- g. when viewed from above, fully obscure Front Wing Strake forward of $X_F = -715$, if fitted.

C3.10.10 Front Wing Adjuster System

The **“Front Wing Adjuster System”** is defined as follows:

- a. **“FW Primary Flap”**, if fitted, must:
 - i. be a continuous portion of Front Wing Profiles, excluding the forwardmost volume.
 - ii. rotate about the **“Primary Axis”**, which must pass through Point A and Point B and be fixed relative to the forwardmost volume of Front Wing Profiles.
- b. **“FW Secondary Flap”**, if fitted, must:

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- i. be a continuous portion of only the rearmost volume of Front Wing Profiles.
 - ii. rotate about the **“Secondary Axis”**, which must pass through Point C and Point D and be fixed relative to either the Primary Flap if fitted, or the forwardmost volume of Front Wing Profiles.
- c. Both FW Primary Flap and FW Secondary Flap must include any fitted Gurneys and any parts of Front Wing Auxiliary Components that are rigidly attached to the respective volumes.
- d. **“Point A”** and **“Point B”** must lie:
 - i. within the volume of FW Primary Flap.
 - ii. no more than 25mm from the forwardmost point of FW Primary Flap at their respective Y positions.
- e. **“Point C”** and **“Point D”** must lie:
 - i. within the volume of FW Secondary Flap.
 - ii. no more than 25mm from the forwardmost point of FW Secondary Flap at their respective Y positions.
- f. Point A and Point C must lie inboard of Y = 150.
- g. Point B and Point D must lie between Y = 475 and Y = 675.

With Front Wing Profiles positioned to comply with Article C3.10.1, up to two pairs of rotation surfaces (four in total) must be defined as follows:

- h. **“Primary Rotation Surfaces”**, which must:
 - i. use the Primary Axis as their axis of revolution.
 - ii. pass through Point A for the first surface and Point B for the second surface.
 - iii. extend over the complete chord of FW Primary Flap.
- i. **“Secondary Rotation Surfaces”**, which must:
 - i. use the Secondary Axis as their axis of revolution.
 - ii. pass through Point C for the third surface and Point D for the fourth surface.
 - iii. extend over the complete chord of FW Secondary Flap.
- j. Over their intersection with the FW Primary and Secondary Flaps at their original design position (defined in accordance with Article C3.10.1), no surface of rotation may have a normal which forms an angle greater than 30 degrees to the Y-Axis.

Compared to the position of Front Wing Profiles defined in Article C3.10.1, adjustment not controlled by the FIA Standard ECU is only allowed when the car is stationary and by the use of a tool, and in accordance with the Sporting Regulation and must:

- k. not result in a decrease in incidence of the tangent lines defined in Article C3.10.1 (g) (i).
- l. result in a maximum deviation for any point of Front Wing Profiles no greater than:
 - i. 30mm for FW Primary Flap.

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- ii. 60mm for FW Secondary Flap.

Any adjustment of Front Wing Profiles controlled by the FIA Standard ECU must:

- m. be about one of the Primary Axis or Secondary Axis only.
- n. when commanded, switch to one of two fixed positions defined as follows:
 - i. a **“Corner Mode”** position, that conforms to a position defined in (k) and that remains identical following any Straight Mode operation to its position beforehand.
 - ii. a **“Straight Mode”** position that, when compared to the Corner Mode position, results in a decrease in incidence of FW Primary Flap and/or FW Secondary Flap. Furthermore, except when limited by a physical stop defined in (u), the magnitude of decrease must remain constant.
- o. have a maximum transition time between the two fixed positions that does not exceed 400ms, measured from the instant at which the command to change mode is issued by the FIA Standard ECU until the position sensor(s), connected to the FIA Standard ECU, confirm that the commanded fixed position has been reached.
- p. be driven by up to two actuators, across both sides of the car.
- q. be measured by position sensors, one for each of the actuators defined in (p), which:
 - i. are mechanically linked to Front Wing Profiles.
 - ii. output analogue signals calibrated over the actuator travel.
 - iii. are connected to the FIA Standard ECU as specified by the FIA.
- r. be symmetrical about $Y=0$.

Furthermore:

- s. Irrespective of whether the adjustment is controlled by the FIA Standard ECU, any fitted FW Primary Flap and FW Secondary Flap must:
 - i. except for Front Wing Fishplates, Front Wing Adjuster Fairing and any fitted Gurneys, remain in their entirety within RV-FW-PROFILES.
 - ii. retain the geometric relationship between all elements of FW Primary Flap, except when adjusting about the Secondary Axis only.
 - iii. retain the geometric relationship between all elements of FW Secondary Flap.
 - iv. conform with all bodywork regulations except for **Article C3.2.4 (b), Article C3.10.11 (g) (i) and C3.4.4 (a)**, sub-clauses (b) to (l) of **Article C3.10.1** and sub-clauses (c) (i), (h) (i), (g) (i) and (m) of **C3.10.11**.
- t. Except for a failure of the Front Wing Adjuster System, or during the transition between Corner Mode and Straight Mode, FW Primary Flap and FW Secondary Flap can only have the two positions defined in (n).
- u. Physical Stops must be provided to prevent both FW Primary Flap and FW Secondary Flap from being rotated external to RV-FW-PROFILES.

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- v. The design is such that failure of the system will result in it returning to its Corner Mode position.
- w. FW Primary Flap and FW Secondary Flap may only be in a State of Deployment when the car is stationary or fully inside an *Activation Zone* as defined by Article B7.1.1 (d).

C3.10.11 Front Wing Auxiliary Components

Once Front Wing Assembly has been fully defined, the following “**Front Wing Auxiliary Components**”, which, unless otherwise stated, are exempt from Article C3.2.4, may be fitted:

- a. In order to ensure the necessary level of sealing, up to four sets of “**Front Wing Fishplates**”, each of which must:
 - i. lie in its entirety within 3mm of a Primary or Secondary Rotation Surface.
 - ~~ii. not exceed the size necessary to allow 20mm of overlap between their adjustable and non-adjustable parts, throughout the range of deviation defined in C3.10.10 (l).~~
 - ii. for adjustment of Front Wing Profiles not controlled by the FIA Standard ECU, not exceed the size necessary to allow 20mm of overlap between their adjustable and non-adjustable parts throughout their permitted range of travel.
 - iii. for adjustment of Front Wing Profiles controlled by the FIA Standard ECU, not exceed the size necessary to allow 20mm of overlap between their adjustable and non-adjustable parts in Corner Mode.
- b. Up to eight Hangers, each of which must comply with C3.2.4 (a) and:
 - i. intersect two adjacent volumes of Front Wing Profiles.
 - ii. be no closer than 150mm to any other Hanger that intersects the same two volumes of Front Wing Profiles.
 - iii. lie in its entirety within RV-FW-PROFILES in Corner Mode.
- c. Up to three “**Front Wing Rotation Brackets**”, for each of FW Primary Flap and FW Secondary Flap, each of which must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-HANGER.
 - ii. intersect either the Primary Axis or the Secondary Axis.
 - iii. lie in its entirety within RV-FW-PROFILES in Corner Mode.
- d. A single “**FW SM CL Fairing**”, which must comply with Article C3.2.4 and:
 - i. lie in its entirety within RV-FW-SM-CL-FAIRING.
 - ii. have up to two sections in any Y-Plane.
- e. A single “**FW SM Mid Fairing**”, that may only be fitted to enclose the actuator defined in Article C3.10.10 (p), and must:
 - i. lie in its entirety within both an axis aligned cuboid of the following dimensions [X, Y, Z] = [250, 50, 100] and RV-FW-SM-MID.
 - ii. comply with Article C3.2.4.

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- iii. be no less than 25mm thick (being the minimum distance when measured normal to the surface).
- iv. contain no radius of curvature less than 5mm.
- f. A single “**FW SM MID Pillar**”, that may only be fitted alongside FW SM Mid Fairing, and must:
 - i. lie in its entirety within both an axis aligned cuboid of the following dimensions [X, Y, Z] = [250, 50, 100] and RV-FW-SM-MID.
 - ii. comply with Article C3.2.4.
 - iii. be arranged such that it is not visible from above with FW SM Mid Fairing in place.
- g. A single “**FW SM Linkage**” which must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-FW-SM-LINKAGE.
 - ii. intersect both Front Wing Profiles and one of Nose or FW SM Mid Fairing.
 - iii. if inboard of Y = 200, be fully obscured from above with Nose present.
- h. A single “**Front Wing Adjuster Fairing**” which must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-FW-ADJUSTER.
 - ii. intersect for the entire range of adjustment, the adjustable part of Front Wing Profiles not controlled by the FIA Standard ECU and the part of Front Wing Profiles that is either stationary or controlled by the FIA Standard ECU.
- i. A single “**Tyre Temperature Fairing**” which must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-FW-SENSOR.
 - ii. intersect either Front Wing Profiles or Front Wing Endplate.
 - iii. lie in its entirety below Z = 350.

Furthermore, once all Front Wing Auxiliary Components have been defined:

- j. they must be Trimmed and Combined with themselves, Front Wing Assembly or Nose.
- k. a Fillet Radius no greater than 10mm, may be applied along the intersection between FW SM CL Fairing and Nose. Fillet radii, no greater than 4mm, may be applied along the intersections between any Front Wing Auxiliary Component and Front Wing Assembly.
- l. the total of Hangers plus Front Wing Rotation Brackets fitted to Front Wing Assembly must not exceed ten.
- m. except for FW SM CL Fairing and Front Wing Fishplates, the Front Wing Auxiliary Components must be arranged such that they are not visible from below with Front Wing Assembly in place.
- n. flexible seals may be fitted between the adjustable and non-adjustable parts of Front Wing Assembly, and Front Wing Fishplates, ~~covering the full range of deviation defined in Article C3.10.10.1.~~
- o. Hangers and Front Wing Rotation Brackets fitted to Front Wing Assembly must lie in their entirety inboard of Front Wing Endplate Body.

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C3.10.12 Front Wing Bodywork Group

Once the components defined in Articles C3.10.1 to C3.10.11 have been constructed in accordance with these provisions, including any subassembly operations, the resulting union is defined as **“Front Wing Bodywork”**.

C3.11 Rear Wing (RW)

C3.11.1 Rear Wing Profiles

“Rear Wing Profiles” Bodywork must:

- a. lie in its entirety within RV-RW-PROFILES.
- b. comprise of up to three, non-intersecting, simply connected volumes.
- c. when assessing each volume independently, contain no more than two sections in any Z-Plane and a single section in any X and Y -Plane.
- d. in any Y-Plane, within 40mm of the rearmost point of the forwardmost section, a line tangent to any part of the section visible from below, must not form an angle greater than 10 degrees when measured against the X-axis.
- e. in any Y-Plane, contain:
 - i. no concave radius of curvature visible from below.
 - ii. no concave radius of curvature less than 100mm visible from above.
 - iii. no concave radius of curvature less than 100mm on the rearmost section only.

Furthermore:

- f. when measured against a Y-Plane, contain no normal to any point on the surface that subtends an angle greater than 20°, or greater than 45° for the region within RV-RW-NOTCH. For volumes thicker than 0.5mm in their entirety (measured normal to the surface on any Y-Plane), compliance with this Article may be determined prior to application of any Edge Radii necessary to demonstrate compliance with C3.3.1 (a).
 - i. compliance with (f) is not required for surfaces that lie exactly on Y = 0 or Y= 575.
- g. at all points along the span, the minimum distance between adjacent Rear Wing Profiles volumes must be between 8mm and 12mm and will be measured with a spherical gauge.

Once the Rear Wing Profiles bodywork is fully defined, a Gurney of up to 10mm may be fitted to the trailing edge of the upper surface of the rearmost section. This Gurney is considered part of the Rear Wing Profiles and must satisfy the provisions of this Article except for (c), (e) and (f).

C3.11.2 Rear Wing Endplate Body

“Rear Wing Endplate Body” Bodywork must:

- a. lie in its entirety within RV-RWEP-BODY.
- b. have up to two sections in any Y-Plane
- c. when viewed from the side, fully obscure RS-RWEP.

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Rear Wing Endplate Body Aerodynamic Surfaces must:

- d. contain no concave radius of curvature less than 100mm.

C3.11.3 Rear Wing Brace

“Rear Wing Brace” Bodywork must:

- a. lie in its entirety within:
 - i. an axis aligned cuboid of the following dimensions [X, Y, Z] = [100, 375, 40].
 - ii. RV-RW-BRACE.
- b. in any Y Plane:
 - i. have at least one axis of symmetry where the larger one will be called the ‘major axis’.
 - ii. have no dimension which exceeds 5mm larger than the major axis.
- c. contain no concave radius of curvature less than 100mm.

C3.11.4 Rear Wing Pylon

“Rear Wing Pylon” Bodywork must:

- a. lie in its entirety within RV-RW-PYLON.
- b. in any Z-Plane:
 - i. have a total area no greater than 5000 mm², except for regions within 30mm of Exhaust Tailpipe.
 - ii. have a section thickness less than 25mm when measured in the Y direction, except for regions inside of RV-TAILPIPE.

Furthermore:

- c. no part of Rear Wing Profiles may be obscured by Rear Wing Pylon when viewed from above.

C3.11.5 Rear Wing Assembly

“Rear Wing Assembly” results from the Trim and Combination of the following components:

- a. Rear Wing Profiles.
- b. Rear Wing Endplate Body.
- c. Rear Wing Brace.
- d. Rear Wing Pylon.

Before trimming is applied, any Rear Wing Profiles and Rear Wing Brace remaining outboard of Rear Wing Endplate Body must be discarded.

A Fillet Radius, no greater than 10mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Rear Wing Assembly must:

- e. be a single volume, with no overlapping regions.

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C3.11.6 Rear Wing Adjuster System

The “**Rear Wing Adjuster System**” is defined as follows:

- a. “**RW Flap**” is constructed solely from Rear Wing Profiles and must:
 - i. exclude all the forwardmost volume.
 - ii. exclude all remaining volumes outboard of $Y = 535$.
 - iii. include all remaining volumes inboard of $Y = 530$.
 - iv. include any fitted Gurney and any parts of Rear Wing Auxiliary Components that are rigidly attached to any included volumes.
- b. Adjustment of RW Flap is about a fixed axis of rotation, which must be aligned with the Y-Axis. Furthermore, in Corner Mode, the axis of rotation must:
 - i. lie within RV-RW-PROFILES.
 - ii. at $Y = 50$, lie between the rearmost point and the mid-point of RW Flap, measured in the X-direction.
 - iii. between $Y = 50$ and $Y = 530$, when viewed from below, be fully obscured by RW Flap.

Any adjustment of RW Flap may only be controlled by the FIA Standard ECU and must:

- c. when commanded, switch to one of two fixed positions defined as follows:
 - i. a “**Corner Mode**” position, that exactly conforms to the position of Rear Wing Profiles defined in Article C3.11.1 and that remains identical following any Straight Mode operation to its position beforehand.
 - ii. a “**Straight Mode**” position that, when compared to the Corner Mode position, results in a decrease in incidence of RW Flap. Furthermore, the magnitude of decrease must always remain identical.
- d. have a maximum transition time between the two fixed positions that does not exceed 400ms, measured from the instant at which the command to change mode is issued by the FIA Standard ECU until the position sensor, connected to the FIA Standard ECU, confirms that the commanded fixed position has been reached.
- e. be driven by a single actuator.
- f. be measured by a position sensor which:
 - i. is mechanically linked to the RW Flap.
 - ii. outputs analogue signals calibrated over the actuator travel.
 - iii. is connected to the FIA Standard ECU as specified by the FIA.

Furthermore:

- g. Any adjustment of RW Flap must maintain:
 - i. the geometric relationship between all parts of RW Flap.

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- ii. compliance with all bodywork regulations except C3.2.4 (b), C3.4.4 (a), Article C3.11.1 and C3.11.7.
- h. Except for a failure of the Rear Wing Adjuster System, or during the transition between Corner Mode and Straight Mode, RW Flap can only have the two positions defined in (c).
- i. The design is such that failure of the system will result in RW Flap returning to its Corner Mode position.
- j. RW Flap may only be in a State of Deployment when the car is stationary or fully inside an *Activation Zone* as defined by Article B7.1.1 (d).
- k. Physical stops must be provided to prevent RW Flap from being rotated beyond these limits.

C3.11.7 Rear Wing Auxiliary Components

Once Rear Wing Assembly has been fully defined, the following “**Rear Wing Auxiliary Components**”, which are exempt from Article C3.2.4, may be fitted:

- a. When RW Flap has two volumes, up to four Hangers, each of which must comply with C3.2.4 (a) and:
 - i. intersect both volumes of RW Flap.
 - ii. be no closer than 140mm to any other Hanger.
 - iii. lie in its entirety within RV-RW-PROFILES when RW Flap is in Corner Mode.
- b. However, between three and five pairs of “**Rear Wing Separators**” must be fitted across both sides of the car. Each pair must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-RW-SEPARATOR.
 - ii. intersect both the forwardmost volume of Rear Wing Profiles and RW Flap.
 - iii. be no closer than 200mm to any other Rear Wing Separator.
 - iv. be designed and arranged such that the relationship between volumes of Rear Wing Profiles can only change whilst the car is in motion in accordance to Article C3.11.6.
 - v. be aligned to provide a bearing across at least 40mm² when the distance between Rear Wing Profiles and RW Flap is at its closest position.
- c. A single “**Rear Wing SM Adjuster Fairing**”, across both sides of the car, which must lie in its entirety within RV-RW-SM-FAIRING.
- d. Up to three “**Rear Wing Rotation Brackets**”, across both sides of the car, where each must:
 - i. lie in its entirety within an individual, freely positioned instance of RV-RW-BRACKET.
 - ii. lie inboard of Y = 20 or outboard of Y = 520.
 - iii. if outboard of Y = 520, intersect both the fixed volumes of Rear Wing Profiles and RW Flap for the entire range of adjustment.
 - iv. lie in its entirety within RV-RW-PROFILES when RW Flap is in Corner Mode.
- e. A single “**Rear Wing Pylon Brace**”, across both sides of the car, which must:

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- i. lie in its entirety between $Z = 400$ and $Z = 590$ and forward of $X_R = 390$.
- ii. not be visible from the side with the Rear Wing Pylons present.
- iii. in any Y Plane, have at least one axis of symmetry where the larger one will be called the 'major axis' and must be less than 100mm in length.
- iv. in any Y Plane, have no dimension which exceeds 5mm larger than the major axis.
- v. in any Y Plane, be no more than 25mm thick, measured normal to the major axis.

Furthermore, once all Rear Wing Auxiliary Components have been defined:

- f. they must be Trimmed and Combined with Rear Wing Assembly.
- g. a Fillet Radius no greater than 10mm, may be applied along the intersection between Rear Wing SM Adjuster Fairing and Rear Wing Assembly, and Rear Wing Pylon Brace and Rear Wing Assembly. Fillet Radii, no greater than 4mm, may be applied along all remaining intersections with Rear Wing Assembly.
- h. they, except for the Rear Wing Pylon Brace, must be arranged such that they are not visible from below with Rear Wing Assembly in place.
- i. flexible seals may be fitted between the adjustable and non-adjustable parts of Rear Wing Assembly and Rear Wing Rotation Brackets, for sealing gaps between these components whilst in Corner Mode.

C3.11.8 Rear Wing Bodywork Group

Once the components defined in Articles C3.11.1 to C3.11.7 have been constructed in accordance with these provisions, including any sub-assembly operations, the resulting union is defined as **"Rear Wing Bodywork"**.

C3.12 Final Assembly

Prior to **"Final Assembly"**, all bodywork groups described in these Articles must be fully defined. Furthermore, each proceeding Final Assembly sub-section must be complete before moving onto the next.

C3.12.1 Front Bodywork to Rear Bodywork Assembly

The Front Bodywork and the Rear Bodywork must be trimmed to each other. The result of this assembly will be known as **"Upper Bodywork"**. A Fillet Radius, no greater than 50mm, may be applied along the intersections between these volumes.

C3.12.2 Upper Bodywork to Floor Bodywork Assembly

The Upper Bodywork and Floor Bodywork must be trimmed to each other. Furthermore:

- a. before trimming, any Engine Cover remaining below the Floor Bodywork must be discarded.
- b. the intersection formed by any remaining Upper Bodywork regions and Floor Bodywork must produce no more than one curve.
- c. a Fillet Radius, no greater than 50mm, may be applied along the intersection between these volumes.

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- d. once all volumes are trimmed and filleted, no part of Engine Cover, the Fillet Radius between Engine Cover and Floor Bodywork, may be visible from below.

C3.12.3 Tail Bodywork to Floor and Upper Bodywork Assembly

The Tail and the Upper Bodywork to Floor Assembly must be trimmed to each other. A Fillet Radius, no greater than 25mm, may be applied along the intersections between these volumes.

C3.12.4 Front Wing Bodywork to Nose Bodywork Assembly

The Front Wing Bodywork and the Nose bodywork must be trimmed to each other. Furthermore:

- a. a Fillet Radius, no greater than 25mm, may be applied along the intersections between these volumes.
- b. except for Front Wing Pylon, FW SM Linkage and FW SM CL Fairing, no part of Front Wing Bodywork may intersect Nose bodywork.

C3.12.5 Rear Wing Bodywork to Tail Bodywork Assembly

The Rear Wing Bodywork and the Tail Bodywork must be trimmed to each other. A Fillet Radius, no greater than 10mm, may be applied along the intersections between these volumes.

C3.13 Bodywork not defined in Articles C3.5 to C3.12

In addition to the Bodywork defined and regulated by Articles C3.5 to C3.12, the following components are permitted:

- C3.13.1** A transparent windscreen, that measures less than 30mm in Z, less than 300mm in Y, and is no more than 3mm thick, may be fixed to the forward face of the Cockpit opening and may extend above RV-CH-MID.
- C3.13.2** Antennae and pitot tubes may be mounted on the upper surface of the Survival Cell ahead of the Cockpit opening and may extend above RV-BODY-FRONT.
- C3.13.3** A fairing may be attached to the Secondary Roll Structure, or the cameras defined in Articles C8.9.3 and C8.16.6. This fairing:
 - a. must lie in RV-HALO.
 - b. must contain no convex radius less than 2mm.
 - c. may be joined to Front Bodywork with a Fillet Radius no greater than 10mm.
- C3.13.4** Ducts (as specified in C3.1.a.ii and C3.1.a.iii) and Primary Heat Exchangers, which must not be visible when viewed from the outside of the car, at any angle perpendicular to the X-axis. This is assessed with the bodywork defined in Articles C3.5 to C3.12 present but prior to the application of Apertures permitted in Articles C3.16.
- C3.13.5** A slip sensor and its fairing may be mounted underneath the Forward Chassis provided it lies within RV-SLIP. The surface of the combined slip sensor and fairings in contact with the External Air Stream must form a single curve when intersected by any Z plane.

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C3.14 Wheel Components

C3.14.1 General Principles

The following principles apply throughout Articles C3.14 and C3.15.

- a. Any Bodywork subject to Wheel Bodywork Assembly as per Article C3.15 must first be classified under one of the groups specified in Article C3.14.
- b. Unless otherwise specified, all Bodywork in Article C3.14 must be present and fitted after Wheel Bodywork Assembly as per Article C3.15.
- c. Compliance of each Bodywork component defined in Article C3.14 and Article C3.15 with Article C3 will be assessed independently and before any Trim and Combination operations and Fillet application described in Article C3.15, and any Aperture application described in Article C3.16.
- d. Compliance of the Aerodynamic Surfaces of each Bodywork component in Article C3.14 will be assessed independently and after their boundaries have been defined by Trim and Combination operations and Fillet application described in Article C3.15, and before any Aperture application described in Article C3.16.
- e. When referred to individually, Wheel Components will have the words “Front” or “Rear” prefixed to the Component name.
- f. The Wheel Component name on its own refers to both the Front and Rear Components simultaneously.

Furthermore, except for permitted Aerodynamic Seals, Wheel Bodywork must:

- g. be rigid and rigidly secured to the suspension uprights.
- h. not be rigidly secured to the suspension members.

Rigidly secured means not having any degree of freedom.

C3.14.2 Drum

“Front Drum” and “Rear Drum” Bodywork, the latter being exempt from Article C3.2.4 (b), must:

- a. be made to the geometry defined by RS-FWH-DRUM and RS-RWH-DRUM respectively.

Furthermore:

- b. Drum Bodywork must be fitted with an Aerodynamic Seal, in the outboard of the two annotated volumes, between the Drum and the axle. This Aerodynamic Seal must be circumferential, continuous (around an arc of 360°), concentric about the Y_w axis, and uniform.
- c. Drum Bodywork may be fitted with a flexible seal or seals, in the inboard of the two annotated volumes, between the Drum and the Wheel Rim.

C3.14.3 Lip

“Front Lip” and “Rear Lip” Bodywork must:

- a. lie in their entirety within RV-FWH-LIP and RV-RWH-LIP respectively.
- b. have up to two sections in any X_w and Y_w -Plane.

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- c. be fully visible from either inboard or outboard, such that any surface obscured from one direction is visible from the other.

Furthermore:

- d. Front Lip Bodywork is optionally fitted and exempt from Article C3.14.1 (b).
- e. Rear Lip Bodywork must have up to two sections in any Z_W -Plane below $Z_W = -170$.
- f. compliance with (c) is not required within regions obscured by RS-RWH-DRUM when viewed from outboard.

Lip Aerodynamic Surfaces must:

- g. contain no concave radius of curvature less than 20mm.
- h. contain no convex radius of curvature less than 20mm.

Furthermore:

- i. compliance with (h) is not required within 10mm of the side-view boundary of the Lip when viewed from inboard or outboard.
- j. compliance with (g) and (h) is not required on the Rear Lip within a single transition between one and two Z_W sections that is contained within two Z_W -Planes up to 10mm apart.

C3.14.4 Scoop

The “**Front Scoop**” and “**Rear Scoop**” Bodywork must:

- a. lie in their entirety within RV-FWH-SCO and RV-RWH-SCO respectively.

Furthermore, Rear Scoop Bodywork rearwards of $X_W = 0$:

- b. is exempt from Article C3.2.4 (b).

Scoop Aerodynamic Surfaces must:

- c. be fully visible from inboard.
- d. contain no radius of curvature less than 20mm, except for regions of the Rear Scoop rearwards of $X_W = 0$.

C3.14.5 Rear Drum Deflector

“**Rear Drum Deflector**” Bodywork must:

- a. lie in its entirety within RV-RWH-FDEF.
- b. have up to two sections in any X_W , Y_W and Z_W -Plane below $Z_W = -170$.
- c. in any Z_W -Plane:
 - i. the rearmost section must contain at least two points 50mm apart.
 - ii. within 50mm of the rearmost point of the section, a line tangent to any part of the section visible from inboard must not form an angle greater than 5 degrees when measured against the X_W -axis.

Furthermore:

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- d. Rear Drum Deflector Bodywork is optionally fitted and exempt from Article C3.14.1 (b).

Rear Drum Deflector Aerodynamic Surfaces must:

- e. contain no concave radius of curvature less than 50mm.

Furthermore:

- f. compliance with (e) is not required within a single transition between one and two Z_W sections that is contained within two Z_W -Planes up to 10mm apart.

C3.14.6 Front Drum Front Deflector

“Front Drum Front Deflector” Bodywork, if present, must:

- a. lie in its entirety:
 - i. within an axis aligned cuboid of the following dimensions $[X_W, Y_W, Z_W] = [150, 45, 25]$, which may be freely translated along any axis, and rotated about the Y_W axis only.
 - ii. below $Z_W = 0$.
- b. comprise of up to two, non-intersecting, simply connected volumes.
- c. have up to two sections in any X_W , Y_W or Z_W -Plane.
- d. not be visible from inboard with the Front Lip present.

Furthermore,

- e. Front Drum Front Deflector Bodywork is optionally fitted and exempt from Article C3.14.1 (b).

C3.14.7 Front Drum Rear Deflector

“Front Drum Rear Deflector” Bodywork, which is exempt from Article C3.2.4, results from the Trim and Combination of the following components:

- a. RS-FWH-RDEF.
- b. Deflector Leading Edge, which must:
 - i. lie in its entirety within RV-FWH-RDEF.
 - ii. contain no more than a single section in any X_W and Z_W -Plane.
 - iii. obscure RS-FWH-TOP when viewed from inboard.

Front Drum Rear Deflector Bodywork must:

- c. be no more than 15mm thick (being the minimum distance when measured normal to the surface), if visible from inboard.

Furthermore:

- d. Compliance with (c) is not required within 5mm of the side-view boundary of **Front Drum Rear Deflector** when viewed from inboard.

Front Drum Rear Deflector Aerodynamic Surfaces must:

- e. in any X_W -Plane forward of $X_W = 150$, contain no concave radius of curvature:

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- i. less than 75mm visible from outboard.
- ii. visible from inboard.
- f. in any X_W -Plane rearward of $X_W = 150$, contain no concave radius of curvature less than 50mm.

C3.14.8 Drum Deflector Stay

Up to three “**Front Drum Deflector Stays**” may be present, each of which must:

- a. lie in its entirety within a right circular cylinder of diameter no greater than 20mm.
- b. lie no closer than 20mm to any other Front Drum Deflector Stay.
- c. in any plane normal to the axis of the right circular cylinder defined in (a), have at least one axis of symmetry.

“**Rear Drum Deflector Stay**” Bodywork may be present, but must:

- d. lie in its entirety within both:
 - i. RV-RWH-STAY.
 - ii. an axis aligned cuboid of dimensions $[X_W, Y_W, Z_W] = [90, 150, 25]$, which may be freely translated along any axis, and rotated about the Y_W axis only.
- e. in any Y_W -Plane:
 - i. have at least one axis of symmetry where the larger one will be called the ‘major axis’.
 - ii. have no dimension which exceeds 5mm larger than the major axis.

Furthermore, all Drum Deflector Stay Bodywork:

- f. is optionally fitted and exempt from Article C3.14.1 (b).

C3.14.9 Wheel Auxiliary Components

The following “**Wheel Auxiliary Components**”, which are optional and exempt from Article C3.14.1 (b), may be fitted:

- a. Up to three Hangers, each of which must:
 - i. intersect the Front Lip and the union of Front Drum and Front Scoop.
 - ii. be no closer than 100mm to any other Hanger.
 - iii. not be visible from inboard with the Front Drum, Front Lip and Front Scoop present.
 - iv. in any Y_W -Plane, have at least one axis of symmetry where the larger one will be called the ‘major axis’.
 - v. in any Y_W -Plane, have no dimension which exceeds 2mm larger than the major axis.
- b. To protect the Scoop inlet from debris ingestion, a “**Front Debris Guard**” and “**Rear Debris Guard**”, which are exempt from Article C3.2.4, must:
 - i. lie in their entirety within RV-FWH-SCO and the union of RV-RWH-SCO and RV-RWH-LIP respectively.

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- ii. when assessing each Debris Guard individually, form one or more sections, each with a cross-sectional area no greater than 8mm^2 when intersected by any Yw-plane.

C3.15 Wheel Bodywork Assembly

Prior to “**Wheel Bodywork Assembly**”, all Bodywork components described in Article C3.14 must be fully defined. Furthermore, each proceeding Wheel Bodywork Assembly sub-section must be completed before moving on to the next.

C3.15.1 Scoop to Drum

“**Scooped Drum**” Bodywork results from the Trim and Combination of the respective Drum and Scoop. A Fillet Radius no greater than 20mm may be applied at the intersection between these volumes.

C3.15.2 Lip to Scooped Drum

“**Outboard Wheel**” Bodywork results from the Trim and Combination of the Scooped Drum and Lip components. A Fillet Radius no greater than 20mm may be applied at the intersection between these volumes. If Front Scooped Drum and Front Lip do not intersect, Front Outboard Wheel must be considered as the non-intersecting assembly of these components and not the result of a Trim and Combination.

Furthermore:

- a. if present, the Front Drum Front Deflector, defined in Article C3.14.6, must be Trimmed and Combined to the Front Outboard Wheel. Fillet Radii no greater than 5mm may be applied at the intersections between these volumes.
- b. if present, the Hangers, defined in Article C3.14.9 (a), must be Trimmed and Combined to the Front Outboard Wheel. Fillet Radii no greater than 4mm may be applied at the intersections between these volumes.
- c. if present, the Debris Guard, defined in Article C3.14.9 (b), must be Trimmed and Combined to the Outboard Wheel. Fillet Radii no greater than 4mm may be applied at the intersections between these volumes.

C3.15.3 Rear Drum Deflector to Rear Outboard Wheel

The Rear Outboard Wheel, Rear Drum Deflector and Rear Drum Deflector Stay, if present, must be Trimmed and Combined to each other and Fillet Radii no greater than 10mm may be applied at the intersections between these volumes.

After trimming, Rear Drum Deflector Stay, if present, must not be visible from inboard or outboard.

The result of this union will be known as “**Rear Wheel Bodywork Assembly**”, which must form a single, simply connected volume with the Debris Guards removed.

C3.15.4 Front Drum Rear Deflector to Front Outboard Wheel

Each Front Drum Deflector Stay, if present, must be Trimmed and Combined to both the Front Outboard Wheel and Front Drum Rear Deflector and a Fillet Radius no greater than 10mm may be applied at the intersections between these volumes.

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After trimming, Front Drum Deflector Stay, if present, must not be visible from inboard.

The result of this union will be known as Front Wheel Bodywork Assembly.

C3.15.5 Internal Cooling Ducts

“Internal Cooling Ducts” are exempt from Article C3.2.4, but must lie in their entirety within Scooped Drum. Once the Apertures in Article C3.16 have been applied, any flow linking the Scoop Inlet to the Scoop Outlet must:

- a. pass through the plane $Y_w = -50$, except for ducts that cool electrical components.

Furthermore, any flow which enters the Scooped Drum must:

- b. not have any resultant flux across a circular section 155mm in diameter with its centre lying on the Y_w axis in the plane $Y_w = -182$ and $Y_w = -211$ for the Front and Rear respectively.

C3.16 Apertures

Once the Final Assembly and Wheel Bodywork Assembly have been fully defined, the apertures listed in the following table may be applied. **“Apertures”** shall be interpreted as mathematical surfaces bound by their peripheries and can only be defined after their respective Reference Volume has been translated and rotated. Unless otherwise stated, each aperture may only be applied once, and must:

- a. lie in its entirety within the stated Aperture RV. Unless otherwise stated, each individual **“Aperture RV”**:
 - i. is an axis aligned cuboid that is bounded by a single internal diagonal extending from $[0,0,0]$ to the stated co-ordinates.
 - ii. may be freely translated in space.
 - iii. may be rotated about $[0, 0, 0]$ within the prescribed limits. $[X^\circ \text{ Limit}, Y^\circ \text{ Limit}, Z^\circ \text{ Limit}]$. Any rotation must be applied before any translation.

All Aperture RVs are referenced to the Car Coordinate System except for the Wheel Bodywork Aperture RVs which are referenced to the Wheel Coordinate System.

- b. be a simply connected surface.
- c. be fully coincident with the surface of the stated Bodywork component and any permitted Fillet or Edge Radius applied to the stated Bodywork component.
- d. have a surface area no greater than the stated area, per side of the car.
- e. unless otherwise stated, and with only the corresponding Bodywork group and Fillet or Edge Radius defined in (c) present, be fully visible from the given direction, with the Aperture surface assumed to be non-transparent.
- f. respect all other stated criteria.
- g. not overlap any other Aperture, except for C3.16.15.

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- h. except for regions of zero or incidental flow, which the FIA may request be demonstrated, only result in flux through the Aperture in the stated direction, when considered relative to the Bodywork. Furthermore, if no direction is stated, the flux direction will not be assessed.

	(c)	(a.i) / (a.iii)	(d)	(e)	(f)	(h)
C3.16.1 Nose	Nose	[15, 40, 20] [0°, ±10°, 0°]	750	Front	i. Must lie in its entirety forward of $X_F = -1100$.	Influx
C3.16.2 Driver Cooling	Driver Cooling	[20, 100, 20] [0°, ±10°, 0°]	1500	Front		Influx
C3.16.3 FW SM Actuator	Front Wing Bodywork or Nose	[30, 30, 120] [±10°, ±25°, ±10°]	see limb. (f) (i).	None	i. Must not exceed the size necessary to accommodate a 10mm offset of the Swept Bounding Volume of FW SM Linkage across the combined range of deviation defined in C3.10.10 (l) and (n). ii. May be fitted with a flexible seal that does not exceed the minimum size necessary to fully close the Aperture. iii. Up to two instances of this aperture are permitted.	
C3.16.4 FW Adjuster	Front Wing Bodywork or Nose	[20, 20, 20] [0°, 0°, 0°]	200	None	i. Up to two instances of this aperture are permitted.	
C3.16.5 FSU IB	Nose and Forward Chassis	[90, 10, 30] [0°, ±10°, ±10°]	2500	Side	i. Up to two apertures (one each for push/pullrod and trackrod) are permitted. ii. No point on the aperture may be more than 100mm from any other point.	
C3.16.6 F-Scoop Inlet	Front Scoop	[50, 425 155, 200] [0°, 0°, 0°]	15,000	None	i. Except for minimal incidental leakage, all air which enters the Front Scoop Inlet must exit the Front Scoop Outlet.	Influx
C3.16.7 F-Scoop Outlet	Front Scoop	[75, 425 155, 200] [0°, 0°, 0°]	15,000	Rear	i. Must lie in its entirety rearward of $X_W = 100$ and above $Z_W = -50$ and no closer than 180mm from the Y_W axis. ii. Except for minimal incidental leakage, only air which has entered the Front Scoop Inlet shall exit the Front Scoop Outlet.	Efflux
C3.16.8 Floor Auxiliary	Floor Bodywork	[10, 80, 80] [0°, ±20°, 0°]	1500	Front	i. Must lie in its entirety above $Z = 100$ and forward of $X_F = 950$.	Influx
C3.16.9 Sidepod Inlet	Sidepod	[20, 475, 250] [0°, ±80°, ±35°]	80,000	Front	i. Must lie in its entirety above $Z = 275$.	Influx
C3.16.10 Cockpit Entry	Chassis Mid	[850, 200, 200] [0°, 0°, 0°]	175,000	Above	i. Visibility must be assessed with the Secondary Roll Structure removed.	
C3.16.11 EC Louvre Cooling	Engine Cover	RV-BW-APERTURE [0°, 0°, 0°]	150,000	None	i. RS-BW-APERTURE must be fully obscured when viewed through the aperture from above. ii. when viewed from the outside of the car, at any angle perpendicular to the X-Axis, any visible exposed surfaces must lie in their	Efflux

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	(c)	(a.i) / (a.iii)	(d)	(e)	(f)	(h)
					entirety within 50mm of the untrimmed surface. iii. Must not be translated.	
C3.16.12 EC Y25 Cooling	Engine Cover	[1200, 25, 50] [0°, ±20°, 0°]	30 000	Above	i. Must lie in its entirety inboard of $Y = 25$. ii. After the aperture has been applied, RS-EC must remain fully obscured when viewed from the side.	Efflux
C3.16.13 EC Rocket	Engine Cover	[10, 300, 350] [0°, 0°, 0°]	100,000	Rear	i. Must lie in its entirety rearward of $X_R = -55$. ii. May consist of up to two apertures per side of the car, where the total area of both apertures must still respect (d) and both apertures must lie within a single instance of the RV.	Efflux
C3.16.14 R-Tyre Sensor	Floor Bodywork or Engine Cover	[5, 60, 30] [0°, ±75°, ±75°]	1500	None	i. Must be fully obscured from below with Floor Body present. ii. Must be arranged such that the infra-red sensor has an uninterrupted line of sight to the rear tyre.	
C3.16.15 RSU IB	Engine Cover	[200, 50, 140] [±90°, 0°, ±90°]	12 000	None	i. A single aperture is permitted for each suspension member and driveshaft provided the suspension member, driveshaft or its respective fairing intersects the Engine Cover Bodywork at Legality Setup. ii. No point on the aperture may be more than 200mm from any other point. iii. At Legality Setup the Aperture must enclose any parts of the suspension member or driveshaft or their respective fairings (if fitted) that lie forward of $X_R = -55$.	Efflux
C3.16.16 R-Scoop Inlet	Rear Scoop	[50, 100, 175] [0°, 0°, 0°]	10,000	None	i. Except for minimal incidental leakage, all air which enters the Rear Scoop Inlet must exit the Rear Scoop Outlet.	Influx
C3.16.17 R-Scoop Outlet	Rear Scoop	[150, 100, 175] [0°, 0°, 0°]	10,000	Rear	i. Must lie in its entirety rearward of $X_W = 100$ and above $Z_W = -50$. ii. Except for minimal incidental leakage, only air which has entered the Rear Scoop Inlet shall exit the Rear Scoop Outlet.	Efflux
C3.16.18 RIS Outlet	Tail	[15, 60, 170] [0°, 0°, 0°]	3250	Rear	i. The RV must lie with its rearmost face at $X_{DIF} = 760$, upper edge at $Z = 380$ and inner edge at $Y = 0$. ii. Is exempt from C3.16 (b).	Efflux
C3.16.19 RIS Light	Tail	[35, 30, 10] [0°, ±20°, 0°]	1000	Below	i. Must lie in its entirety behind $X_{DIF} = 350$ for the purpose of cooling RIS mounted electronics.	Influx
C3.16.20 Cockpit Cooling	Forward Chassis	[100, 100, 10] [0°, ±10°, 0°]	3000	Below	i. Must lie in its entirety between $X_{A300} < X < X_{C-950}$. ii. All air entering the aperture must exit into the Survival Cell.	Influx

C3.17 Suspension and Driveshaft Fairings

C3.17.1 “Suspension Fairings” must be fitted to all suspension members defined in Article C10.3.6, except for those of circular cross section. Furthermore, Suspension Fairings must be internally sealed to prevent any internal airflow.

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- C3.17.2** Each Suspension Fairing, relative to its corresponding suspension member, must:
- fully cover the member and any components mounted to the member, over the span exposed to the External Air Stream.
 - be rigid and have zero degrees of freedom relative to the member. Except for suspension conforming to Article C10.3.3, where minimal deformation or articulation will be allowed in shared cover sections to accommodate nominal misalignment of relevant suspension members through the suspension travel range.
 - lie entirely ahead of $X_R = 300$.
- C3.17.3** The surface bounded by the external cross-section of each Suspension Fairing, when taken normal to the corresponding suspension member's load line defined in Article C10.3.6 (a) must:
- intersect the load line, except for front suspension elements, locally, for the sole purpose of ensuring minimum clearance with the Wheel Rim at full steering lock.
 - have at least one axis of symmetry where the larger one will be called the "major axis."
 - have no dimension which exceeds:
 - 100mm.
 - 5mm larger than the major axis.
 - have an aspect ratio no greater than 3.5:1, where aspect ratio is the ratio of the major axis to the maximum thickness, measured in the direction normal to the major axis.
 - have an incidence (defined as the angle between the major axis and the plane $Z=0$ with the car at its Legality Setup, as defined in Article C10.1) which lies within the following ranges:
 - For Front Suspension Fairings, between 10deg (nose down) and 0deg.
 - For Rear Suspension Fairings, between 10deg (nose down) and -10deg (nose up).
- C3.17.4** Suspension Fairings of members which share an attachment point will be considered by a virtual dissection into discrete components. Such fairings will be allowed minimally necessary exceptions to Article C3.17.3 (b), (c.ii), (d) and (e) close to the junction of the suspension members, including minimal cut outs to allow the passage of a suspension member connected inboard to the rockers defined in Article C10.4.1. After application of any such cut out, the resultant fairings must still seal the internal volume of the fairings from the External Air Stream.
- C3.17.5** A **"Driveshaft Fairing"** may be fitted around the driveshaft defined in Article C9.10 and must cover the driveshaft and any component mounted to it, over the span exposed to the External Air Stream. Furthermore, this fairing must be internally sealed to prevent any internal airflow.
- The surface bounded by the external cross-section of this fairing, when taken normal to a single nominated legality axis which over the driveshaft's length must lie in its entirety within 25mm of the driveshaft, must:
- respect the criteria of Article C3.17.3 except for:
 - part (a), which must instead intersect the legality axis.
 - part (c.i), which is increased to 150mm.

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- iii. part (e), where the range is between 10deg (nose down) and -10deg (nose up).

Furthermore, this fairing must not vary the angle subtended by the major axis of its cross section when taken normal to the legality axis, and the plane $Z = 0$ by more than $\pm 5\text{deg}$ from that at Legality Setup over the range of suspension movement.

C3.17.6 The angle between the normal to any point of the surface of a Suspension or Driveshaft Fairing and a plane normal to the corresponding suspension member's load line or the driveshaft's legality axis must not exceed 15 degrees. This is except for regions close to the:

- a. inner attachment point.
- b. outer attachment point.
- c. junction between suspension members that share an attachment point.

Treating the Suspension or Driveshaft Fairing as a solid volume bound by its Bodywork surfaces, for fairings thicker than 0.5mm in their entirety (measured normal to the surface), compliance with this Article may be determined prior to application of any Edge Radii necessary to demonstrate compliance with C3.3.1 (a).

C3.17.7 In order to permit suspension travel while maintaining an Aerodynamic Seal, minimal flexible components may be fitted between a Suspension or Driveshaft Fairing and the following components:

- a. Wheel Bodywork, as defined in Article C3.15.
- b. Front Bodywork, as defined in Article C3.7.
- c. Engine Cover Bodywork, as defined in Article C3.8.2.
- d. Tail Bodywork, as defined in Article C3.9.
- e. Gearbox Case, as defined in Appendix C1.
- f. Rear Wing Pylon, as defined in Article C3.11.4.
- g. The Suspension Fairing covering the suspension member connected inboard to the rockers defined in Article C10.4.1.

With the car at its Legality Setup, these flexible components must:

- h. contain a Fillet Radius of up to 30mm between the components they are sealing.
- i. except for the Fillet Radius defined in (h), conform with the shape restrictions of either of the two components they are sealing. For clarity, parts of these components may be replaced with the flexible component.
- j. be designed in such way as to maintain their shape integrity with the car in motion, and as much as possible, with the suspension at another position than that of the Legality Setup.

When the suspension is at Legality Setup and the Wheels are in the straight-ahead position, exceptions to (h) and (i) are permitted in regions internal to the volume bounded by the Fillet Radius defined in (h) between:

- k. Suspension Fairings and Front Wheel Bodywork.

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l. the Front Trackrod Suspension Fairing and Front Bodywork.

C3.17.8 Suspension Fairings may be united with the Driveshaft Fairing to form a solitary fairing. Such a fairing will be considered by a virtual dissection into discrete Suspension or Driveshaft Fairings. Furthermore, the discrete Suspension and Driveshaft Fairings which form the solitary fairing, may have minimally necessary exceptions to Article C3.17.3 (b), (c.ii), (d) and (e), Article C3.17.5 (a.iii) and Article C3.17.6, close to a single junction where these fairings transition from virtually discrete to physically discrete.

C3.18 Aerodynamic Component Flexibility

C3.18.1 Introduction of load/deflection tests

Bodywork subjected to load/deflection tests in this Article must be of rigid construction and exhibit nominally linear elastic behaviour up to the maximum load applied whilst running on track, unless stated otherwise.

Any system, mechanism, or procedure, that achieves, seeks to, or is suspected of, circumventing the tests defined in this Article is prohibited.

In order to ensure that the requirements of Article C3.2.2 are respected, the FIA reserves the right to introduce further load/deflection tests, modify existing tests, or require F1 Teams to provide further evidence as stated in Article C1.7, for any part of the bodywork which appears to be, or is suspected of, moving whilst the car is in motion.

Further details about the implementation of each test are given in [FIA-F1-DOC-050](#).

Note: The format of articles C3.18.2 through C3.18.14 has been changed. For clarity, only substantive changes have been highlighted

C3.18.2 Front Wing Bodywork Flexibility

Field	Description
Component	Front Wing Bodywork
Load(s)	a) Symmetric: 1000 N per car side simultaneously b) Asymmetric: 1000 N
Load Direction(s)	[0, 0, -1]
Load Application Point(s)	[X _F =-925, ±525, 300]
Load Application Method	50 mm diameter ram(s) on F1 team-supplied adaptors
Adaptor / Pad Details	350 × 100 mm Inner face at Y = ±475 Forward face at X _F = -1100 Top face at Z = 300 Flat top surface, no recesses Mass ≤ 2 kg
Displacement Reference	Survival Cell at X _A = 0
Displacement Direction	Along loading axis

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Deflection Limit(s)	a) Symmetric: ≤ 15 mm b) Asymmetric: ≤ 20 mm
Test Conditions	Test a) and b) done separately
Notes	Adaptor must not increase rigidity of the bodywork under test

C3.18.3 Front Wing Flap Flexibility

Field	Description
Component	Front Wing Primary and Secondary Flaps
Load(s)	60 N point load
Load Direction(s)	Normal to flap surface
Load Application Point(s)	Any point on trailing edge
Load Application Method	Point load, pad fully on flap surface
Adaptor / Pad Details	25 mm diameter rubber pad
Displacement Reference	Forward-most Front Wing profile
Displacement Direction	Along loading axis
Deflection Limit(s)	≤ 10 mm
Test Conditions	Flap in Corner Mode; may be combined with C3.18.2
Notes	N/A

C3.18.4 Front Wing Endplate Flexibility

Field	Description
Component	Front Wing Endplate
Load(s)	60 N
Load Direction(s)	a) $[0, 0, -1]$ b) $[0, +1, 0]$ or $[0, -1, 0]$
Load Application Point(s)	Anywhere on endplate periphery
Load Application Method	Point load, pad fully on endplate surface
Adaptor / Pad Details	25 mm diameter rubber pad, fully on surface
Displacement Reference	Forward-most Front Wing profile at $Y = +575$ or $Y = -575$
Displacement Direction	Along loading axis
Deflection Limit(s)	≤ 7 mm
Test Conditions	Test a) and b) done separately
Notes	N/A

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C3.18.5 Front Floor Flexibility

Field	Description
Component	Front Floor Bodywork
Load(s)	$\leq 8000\text{N}$ 6000N
Load Direction(s)	[0, 0, +1]
Load Application Point(s)	25 mm from [$X_F = 500$ 800 , 0, -10]
Load Application Method	10 mm diameter gimble pad
Adaptor / Pad Details	N/A
Displacement Reference	Survival Cell
Displacement Direction	Along loading axis
Load / Deflection Limit(s)	Load envelope to exceed at any deflection given by the straight-line graphs, for load: - Increasing: (0mm, 0N), (6mm, 6000N 5000N), (25 mm, 6000N 5000N) - Decreasing: (25 mm, 6000N 5000N), (6.5mm 7mm, 6000N 5000N), (0.5mm 1mm, 0N)
Test Conditions	N/A
Notes	Test limited to 8000N 6000N or 15mm (whichever occurs first), unless otherwise requested by the FIA Load-deflection relationship must be strictly monotonic during loading and unloading

C3.18.6 Outboard Floor Flexibility

Field	Description
Component	Floor bodywork group
Load(s)	a) 500 N per car side simultaneously b) 60 N
Load Direction(s)	[0, 0, -1]
Load Application Point(s)	a) [$X_R = -525$, $Y = \pm 550$] b) Anywhere along the floor edge
Load Application Method	a) 50 mm diameter ram on F1 team-supplied adaptors b) Pad entirely on the bodywork surface
Adaptor / Pad Details	a) 50 mm diameter adaptors centred on the load axis b) 25 mm diameter rubber pad
Displacement Reference	Rear scrutineering supports defined in C3.3.4
Displacement Direction	Along loading axis
Deflection Limit(s)	a) ≤ 8 mm b) ≤ 7 mm
Test Conditions	Test a) and b) done separately
Notes	N/A

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C3.18.7 Central Floor Flexibility

Field	Description
Component	Central Floor Bodywork (RV-PLANK)
Load(s)	a) Up to 6 kN b) Up to 10 kN
Load Direction(s)	[0, 0, +1]
Load Application Point(s)	a) Hole between $X_c = -600$ and $X_c = -800$ b) Rear plank hole
Load Application Method	Ram on FIA provided adaptor, centred on the plank holes
Adaptor / Pad Details	Contacting the central floor bodywork with 3 × 10mm diameter pads, equispaced on a 45mm pcd, with a gimbed interface to the loadcell
Displacement Reference	Scrutineering supports defined in C3.3.4
Displacement Direction	Along loading axis
Deflection Limit(s)	a) ≥ 3 kN/mm b) ≥ 6 kN/mm
Test Conditions	All rotational orientations of adaptor assembly Test a) and b) done separately
Notes	Stiffness limit must be achieved for any load or displacement

C3.18.8 Floor Board Flexibility

Field	Description
Component	Floor Bodywork
Load(s)	a) 100 N b) 60 N
Load Direction(s)	a) [0, 0, -1] b) [0, +1, 0] or [0, -1, 0]
Load Application Point(s)	a) $[X_F, Y] = [695, \pm 720]$ b) Bodywork periphery
Load Application Method	a) Hanging weights on F1 team-supplied adaptor b) Point load, pad fully on bodywork surface
Adaptor / Pad Details	a) 25 mm diameter adaptors centred on load axis. Holes to accommodate a 2.3 mm rod to pass through b) 25 mm rubber pad
Displacement Reference	Front scrutineering supports defined in C3.3.4
Displacement Direction	Along loading axis
Deflection Limit(s)	a) ≤ 5 mm b) ≤ 7 mm
Test Conditions	Tests a) and b) performed separately
Notes	All required holes must be sealed on circuit.

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C3.18.9 Diffuser Flexibility

Field	Description
Component	a) Diffuser b) Floor Winglets
Load(s)	a) 100 N per car side simultaneously b) 60 N
Load Direction(s)	a) [0, 0, -1] b) Normal to surface
Load Application Point(s)	a) 15 mm forward of rear extremity at $Y = \pm 320$ or ± 215 b) Anywhere
Load Application Method	a) Hanging weights on F1 team-supplied adaptor b) Point load, pad fully on bodywork surface
Adaptor / Pad Details	a) 25 mm diameter adaptors centred on load axis. Holes to accommodate a 2.3 mm rod to pass through b) 25 mm rubber pad
Displacement Reference	RIS
Displacement Direction	Along loading axis
Deflection Limit(s)	a) ≤ 5 mm b) ≤ 7 mm
Test Conditions	Tests a) and b) performed separately
Notes	All required holes must be sealed on circuit on both upper and lower surfaces

C3.18.10 Rear Wing Bodywork Flexibility

Field	Description
Component	Rear Wing Profiles
Load(s)	[218, 0, -985] N per car side simultaneously
Load Direction(s)	As load vector
Load Application Point(s)	$[X_R = 390, \pm 325, 835]$
Load Application Method	50 mm diameter rams on F1 team-supplied adaptors
Adaptor / Pad Details	Adaptors must lie between $Y = \pm [275-375]$ and $X_R = 150-500$ Upper surface at $Z = 850$ with 52 mm counterbore for the ram Must be removable by translation along Z-axis only
Displacement Reference	<u>Displacement</u> : RIS <u>Angular</u> : on the rear wing profiles, at $[X_R, Y] [375, \pm 450]$ and relative to the RIS
Displacement Direction	<u>Displacement</u> : along loading axis <u>Angular</u> : in Y-plane
Deflection Limit(s)	≤ 6 mm linear $\leq 1.0^\circ$ angular
Test Conditions	Both sides loaded simultaneously

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Notes	N/A
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C3.18.11 Rear Wing Flap Flexibility

Field	Description
Component	Rear Wing Flap
Load(s)	500 N
Load Direction(s)	[+1, 0, 0]
Load Application Point(s)	a) [Y, Z] = [Y, 800] with Y ≤ 50 b) [Y, Z] = [+300, 800] c) [Y, Z] = [-300, 800]
Load Application Method	Load on F1 team-supplied adaptors
Adaptor / Pad Details	25 mm wide
Displacement Reference	Forward part of Rear Wing Mainplane at same Y
Displacement Direction	Along loading axis
Deflection Limit(s)	≤ 7 mm
Test Conditions	One load position at a time
Notes	N/A

C3.18.12 Rear Wing Mainplane Trailing Edge

Field	Description
Component	Rear Wing Mainplane Trailing Edge
Load(s)	200 N
Load Direction(s)	Normal to lower surface
Load Application Point(s)	Trailing edge at Y = 0, ±150 or ±450
Load Application Method	Load on F1 team-supplied adaptors
Adaptor / Pad Details	≤ 50 mm wide ≤ 10 mm forward of trailing edge 8 mm female thread in the underside
Displacement Reference	Forward part of Rear Wing Mainplane at same Y-station
Displacement Direction	Along loading axis
Deflection Limit(s)	≤ 3 mm
Test Conditions	One Y location at a time
Notes	N/A

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C3.18.13 Rear Wing Endplate Flexibility

Field	Description
Component	Rear Wing Endplates
Load(s)	50 N
Load Direction(s)	[0, ±1, 0] inward
Load Application Point(s)	[X _R = 670, Z = 550]
Load Application Method	Point load
Adaptor / Pad Details	25 mm rubber pad
Displacement Reference	Opposite rear wing endplate at same X and Z
Displacement Direction	Along loading axis
Deflection Limit(s)	≤ 10 mm
Test Conditions	N/A
Notes	N/A

C3.18.14 Rear Wing Skins

Field	Description
Component	Rear Wing Profile Skins
Load(s)	60 N
Load Direction(s)	Normal to and away from element
Load Application Point(s)	Skin surface
Load Application Method	Vacuum cup
Adaptor / Pad Details	50 mm diameter vacuum cup
Displacement Reference	Lower surface ≥ 300 mm away at same X
Displacement Direction	Normal to surface
Deflection Limit(s)	≤ 2 mm
Test Conditions	N/A
Notes	N/A

C3.18.15 Additional Diagnostics

To allow the behaviour of Bodywork to be monitored, F1 Teams are required, as determined by the FIA, to apply contrasting markers to Bodywork and may be required to run additional cameras provided by the FIA. Details and positions of these markers and additional Cameras are given in the document [FIA-F1-DOC-050](#).

An F1 Team may be required by the FIA to run additional cameras specified by the FIA in Position 2 RHS (C8.16.7) and/or Position 3 (as defined in C8.16) in any free practice session. Such a camera will not be considered as Bodywork.

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C3.19 Aerodynamic Component construction

C3.19.1 Front Wing Endplate

The bodywork declared as Front Wing Endplate in Article C3.10.6 must be made to the laminate PL-ANTISPLINTER.

Fasteners and related inserts, for the purpose of attaching the front wing endplate to the profiles must be positioned at least 30mm rearward of the leading edge of the endplate.

Metallic components are allowed but must not cause a risk to other competitors. The Technical Delegate may ask an F1 Team to modify or remove metallic components that may pose such a risk.

C3.19.2 The Secondary Roll Structure fairings, defined in Article C3.13.3, must be made to the laminate PL-HALO.

C3.19.3 Floor

Metallic components used in the Floor Bodywork must not cause a risk to other competitors.

The Technical Delegate may ask an F1 Team to modify or remove metallic components that may pose such a risk

C3.19.4 Bib

Floor Bodywork or geometry within RV-BIB-STAY that lies within the plan view area of RV-FLOOR-BIB and below $Z = 150$, may incorporate isolated sections of either compliant material or thin laminate to accommodate the motion of the Floor Bodywork, as described in Article C3.18.6, resulting from contact between the plank assembly and the ground.